



# EXECUTIVE REFERENCE DATA NATIONAL ENERGY MANAGEMENT

NATIONAL ENERGY COUNCIL  
2014



# PREFACE

---

In order to provide updated information on energy situations in Indonesia, the National Energy Council deemed necessary to re-publish an improved publication of the 2013 booklet on Executive Reference Data: National Energy Management. This first edition in 2014 presenting information on energy management up to 2013, which includes socio-economic, energy indicator, the management each type of energy covering oil and gas, coal, renewable energy, and electricity. Slightly different from the 2013 edition, this publication provides also information related to energy by sector as well as Indonesia positions in the global context.

The economic growth of Indonesia at an average annual rate of 5.78% during the period 2012 through 2013, has triggered the increase of domestic energy demand by an average of 4.4% per year. By 2030 the energy demand is estimated to be three times the demand level in 2010, while the increase in energy production in 2030 is forecast to rise at twice that of the year 2010. This condition requires a good energy planning and management to ensure continuity of supply of energy in the long term.

We would like to thank and appreciate all parties, for their contribution to the preparation of this booklet. It is hoped that this publication will be of great benefits for energy management in Indonesia and a useful tool to the national and global energy discussions.

Jakarta, July 2014

Secretary General of  
National Energy Council

# TABLE OF CONTENT

---

|                                                     |    |
|-----------------------------------------------------|----|
| 1. Socio-economic.....                              | 3  |
| 2. Energy Indicator.....                            | 9  |
| 3. Oil and Gas.....                                 | 15 |
| 4. Coal.....                                        | 31 |
| 5. Electricity.....                                 | 37 |
| 6. Progress of FTP-1 and FTP-II.....                | 43 |
| 7. New and Renewable Energy.....                    | 47 |
| 8. Energy Consumption by Sector.....                | 53 |
| 9. CO <sub>2</sub> Emissions.....                   | 61 |
| 10. Indonesia's Position in the Global Context..... | 64 |

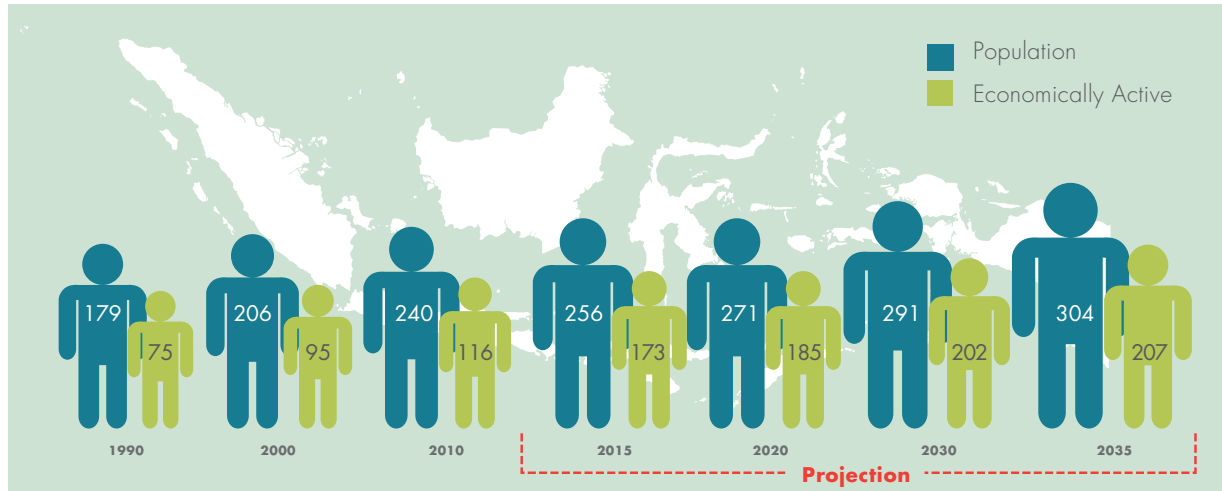


A solid orange horizontal bar spanning the left side of the slide.

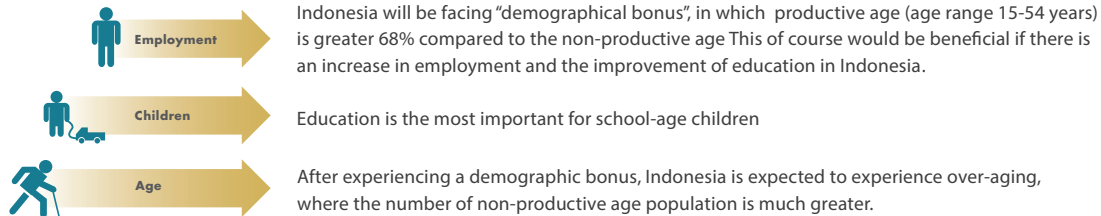
**SOCIO - ECONOMIC**

# POPULATION

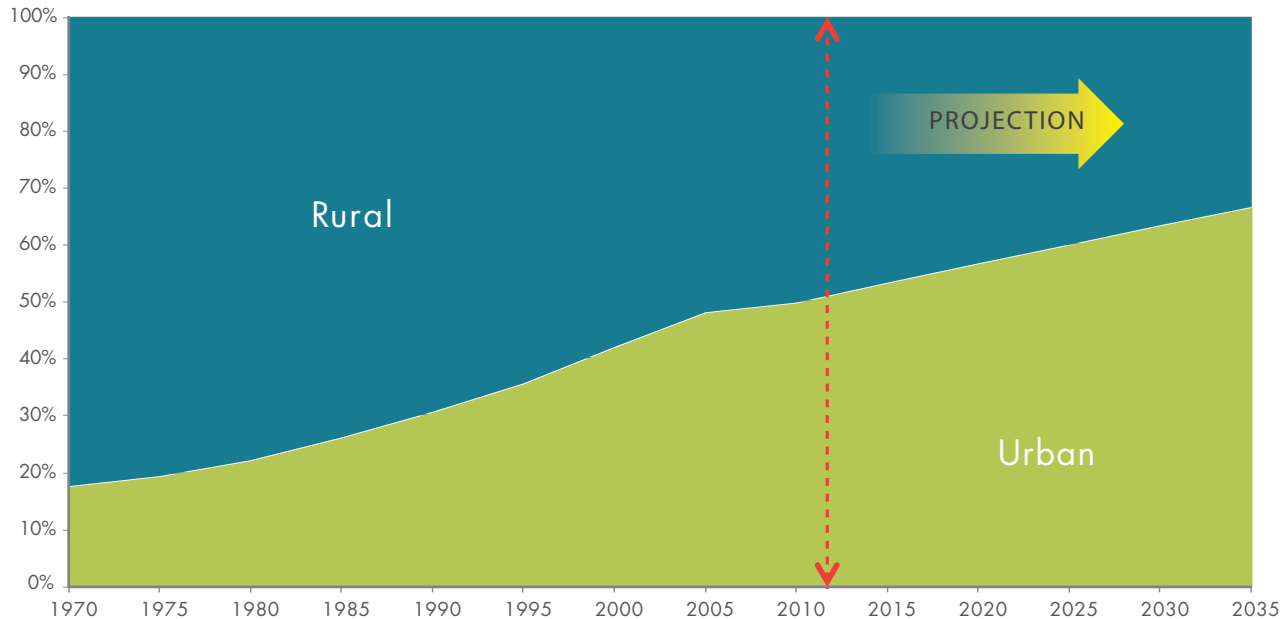
(Million Peoples)



Source: Indonesian Central Bureau of Statistics, reprocessed by NEC



# POPULATION IN RURAL AND URBAN AREAS



Source: Indonesian Central Bureau of Statistics, reprocessed by NEC

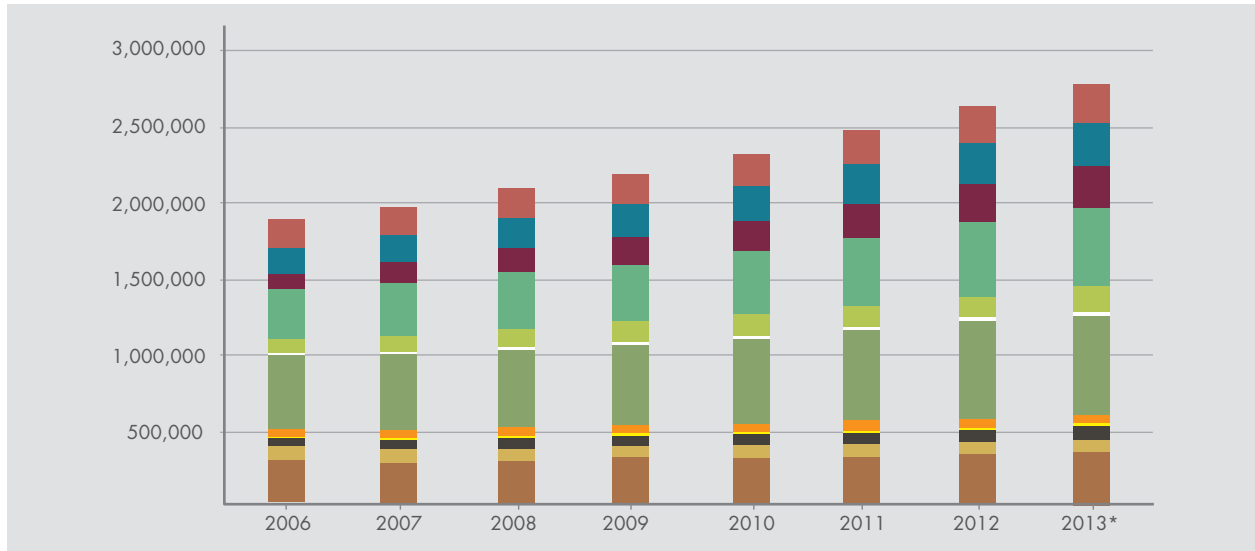
In 1970, number of population in urban area 17.5% increase 48.1% in 2005 (increase an average 2.9 % per year).

In period 2005 – 2013, Urban people increase slowly by an average of 1.2 % per year.

In 2035, the population in the urban areas is estimated/projected to be increased by the composition of 66.6% living in town/city and 33.4% living in villages. The change of a village to become a town has caused the increase in number and density of population, and the economic activity is no longer relies on the agricultural sector, due to the improvement of infrastructure.

# GDP PER SECTOR (CONSTANT 2000)

(Billion IDR)



Source: Indonesian Central Bureau of Statistics, reprocessed by NEC

- Services
- Transport & Communication
- Construction
- Non-Oil & Gas Manf. Industry
- Quarrying
- Oil & Gas Mining
- Finance
- Trade, Hotel & Restaurants
- Electricity, Gas & Water Supply
- Oil & Gas Manf. Industry
- Non-Oil & Gas Mining
- Agriculture

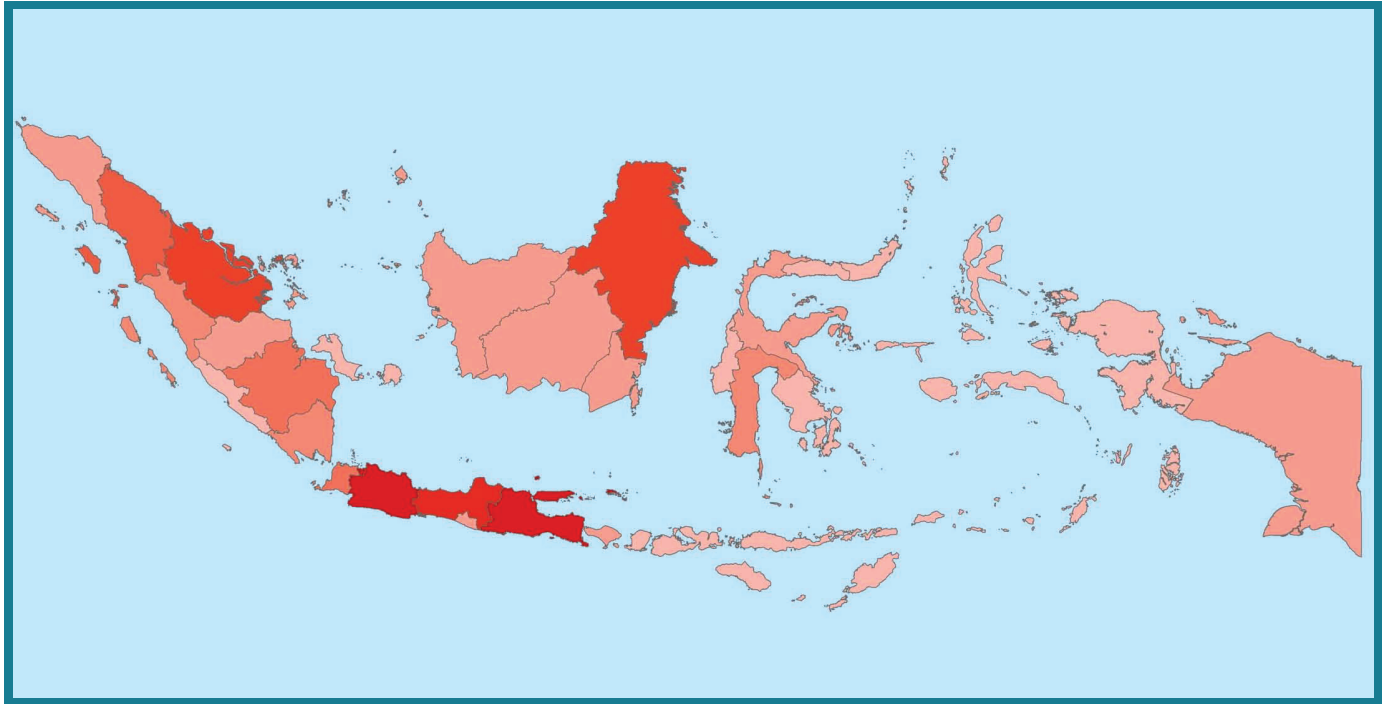
\* Temporary Value

In 2013, GDP of Indonesia is 2,770 trillion rupiahs or increased one and a half times when compared with the year 2006 (growth 5.96 % per year), with the growth occurs in all sectors (the highest growth is in the in transportation and communication sectors about 12.9 % per year). Oil & Gas and Mining industry decreased about 1.1 % and 0.99 % per year.

In 2013, the share of mining, oil and gas Industry to the Indonesia's total GDP is about 3% and 2% respectively.

# GDP BY REGION

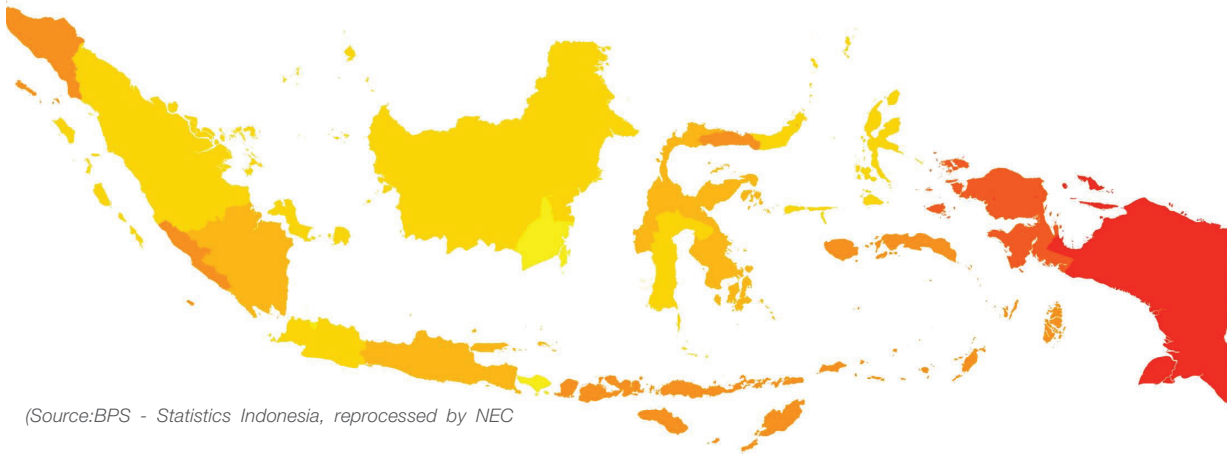
(GDPR in Billion IDR)



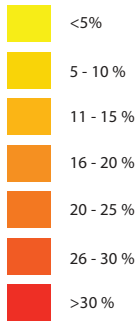
Source: BPS - Statistics Indonesia, reprocessed by NEC

# POOR PEOPLE, 2013

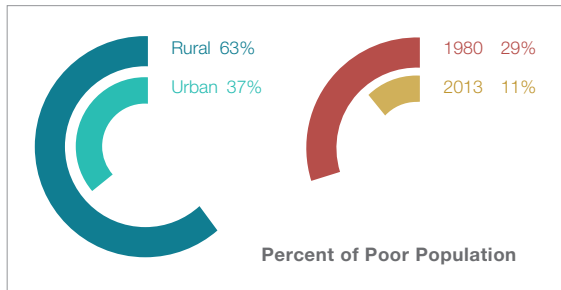
(%)



(Source: BPS - Statistics Indonesia, reprocessed by NEC)



\*) September 2013



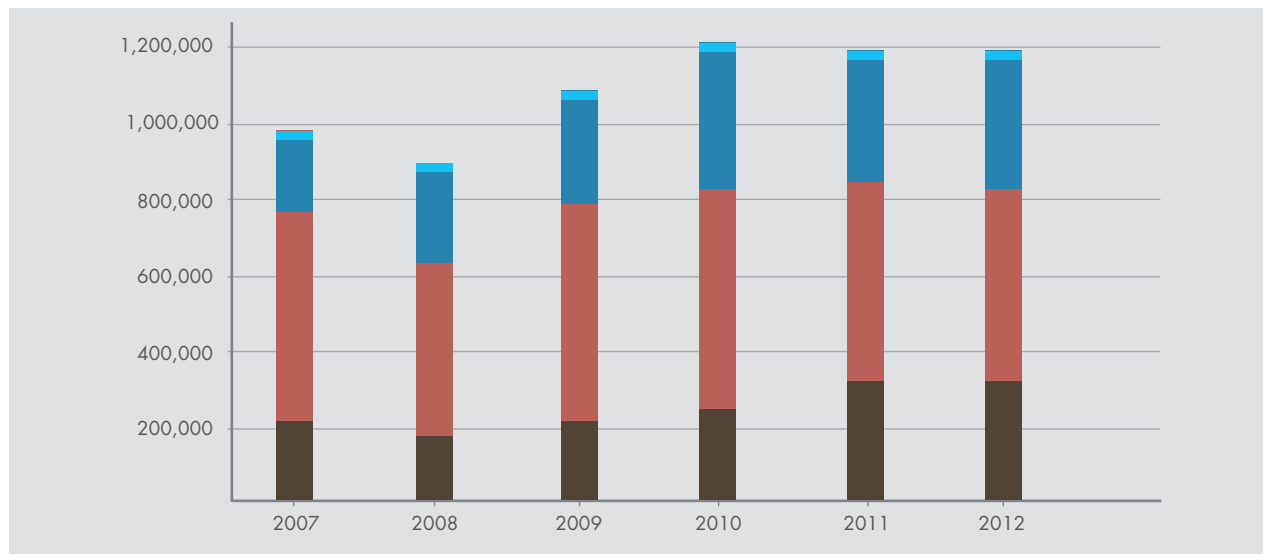
28 million people (11%) of Indonesia's population is poor, of which 8.53% domiciling in the urban area and 14.42% residing in the rural area.



**ENERGY INDICATOR**

# PRIMARY ENERGY SUPPLY

(MBOE)



|           | 2007    | 2008    | 2009    | 2010    | 2011    | 2012    |
|-----------|---------|---------|---------|---------|---------|---------|
| Others RE | 2,873   | 3,193   | 3,991   | 4,434   | 4,668   | 6,025   |
| Hydro     | 28,451  | 29,060  | 28,696  | 44,003  | 30,074  | 31,795  |
| Gas       | 183,624 | 193,352 | 253,198 | 269,942 | 279,286 | 275,950 |
| Oil       | 474,033 | 464,704 | 546,050 | 572,249 | 497,729 | 485,045 |
| Coal      | 258,174 | 205,492 | 236,439 | 281,400 | 334,143 | 344,400 |

Source: Ministry of Energy and Mineral Resources/MEMR, reprocessed by NEC

\*) Excluded Biomass

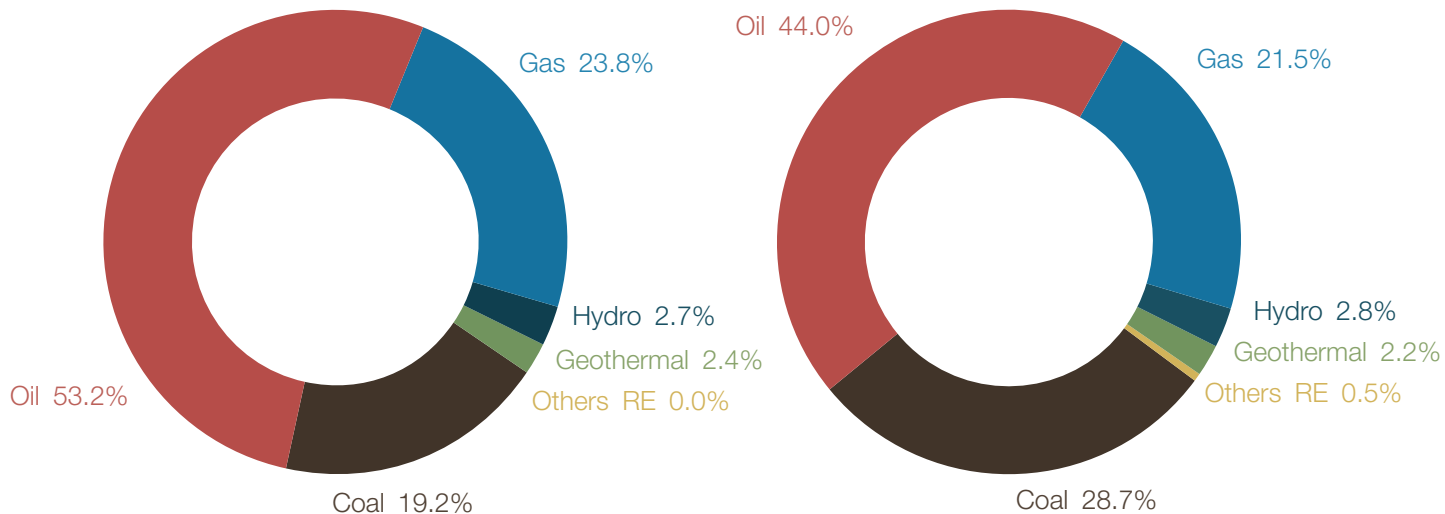


# PRIMARY ENERGY MIX

2003  
858 Million BOE

3.8% p.a.  
Annual growth

2013\*  
1,243 Million BOE

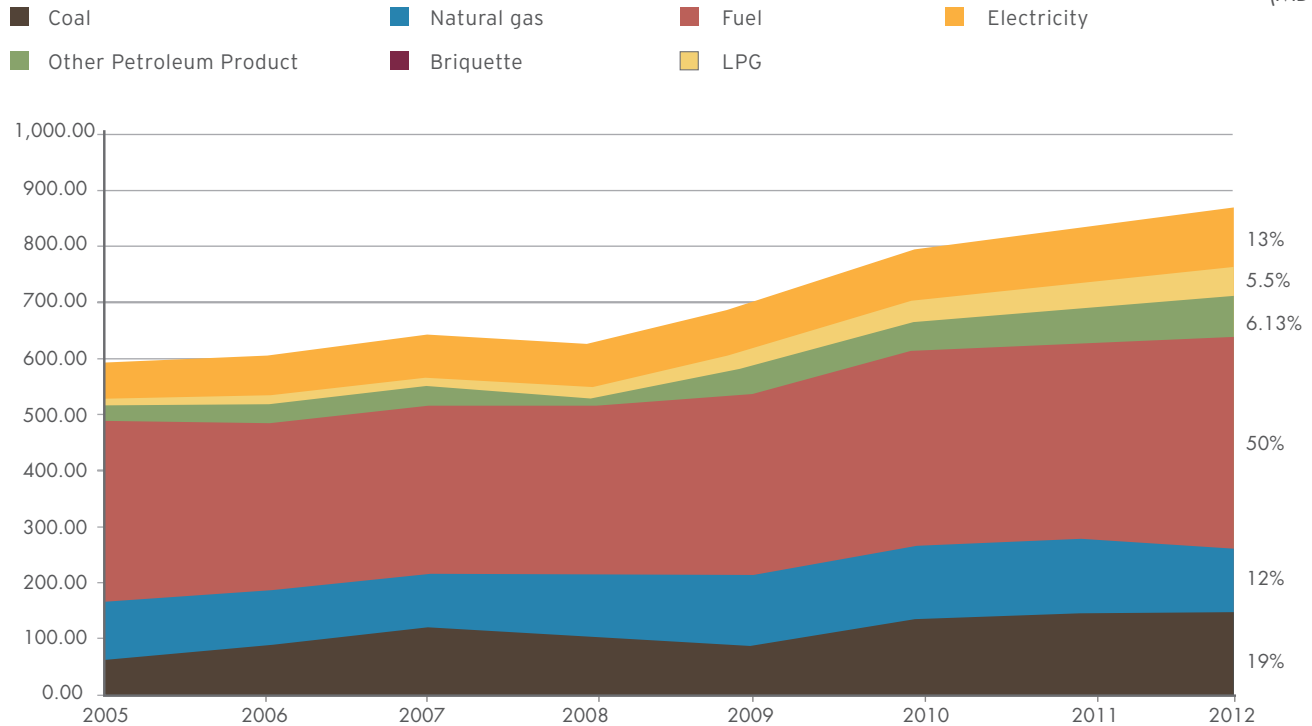


Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

\*) Temporary data

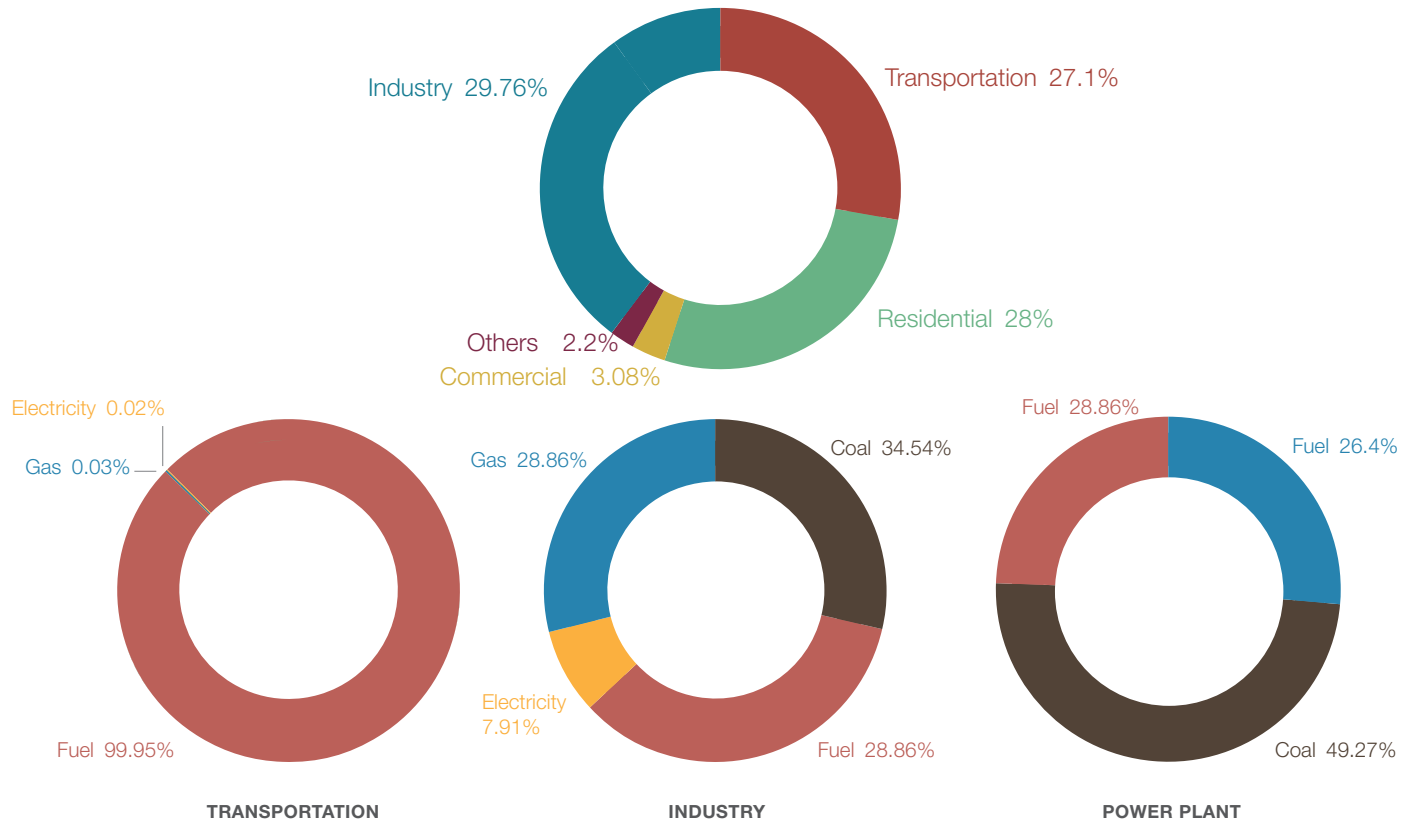
# FINAL ENERGY CONSUMPTION

(MBOE)



Source : BPS - Statistics Indonesia, reprocessed by NEC

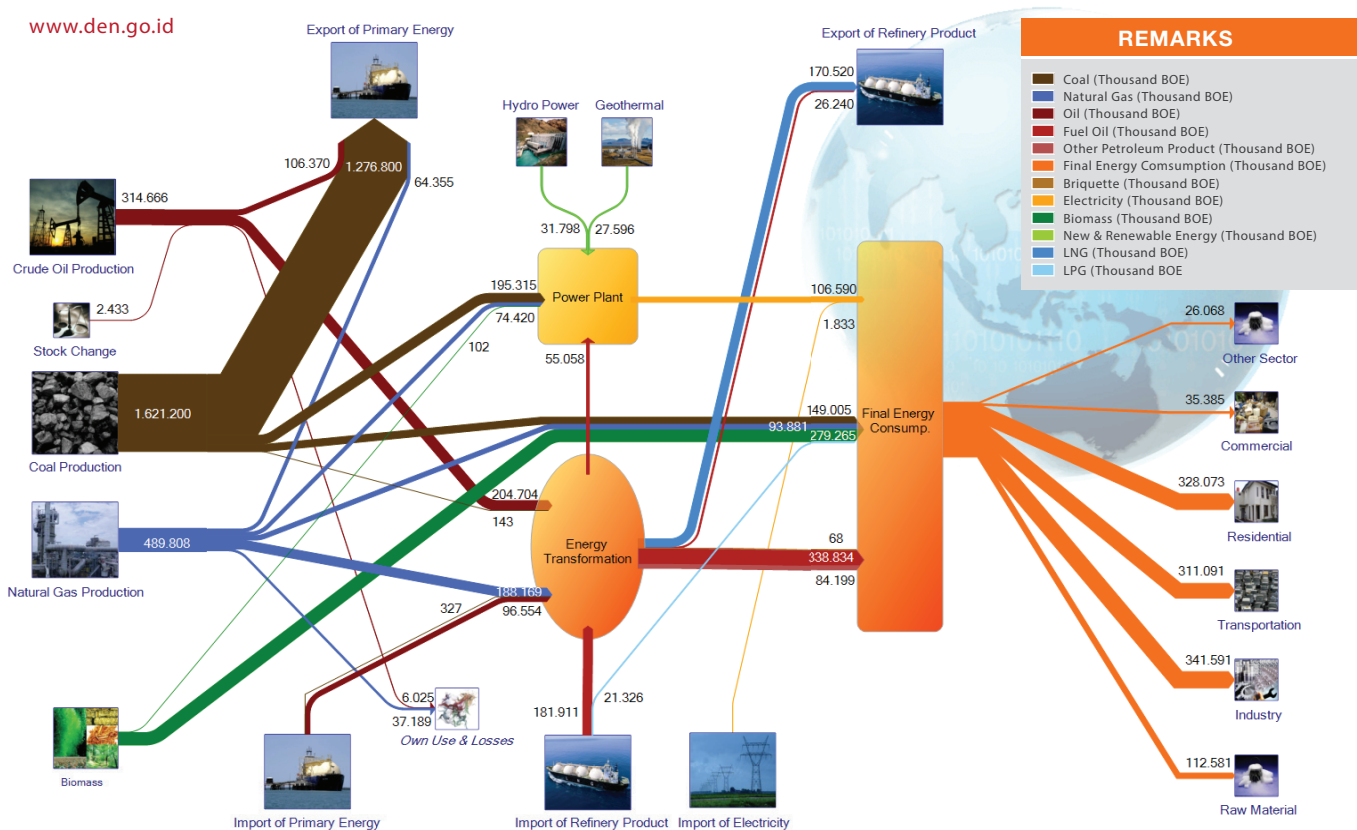
# SHARES OF ENERGY UTILIZATION BY SECTOR



Source : BPS - Statistics Indonesia, reprocessed by NEC

# ENERGY FLOW, 2013\*)

www.den.go.id



A solid orange horizontal bar spanning the width of the page.

**OIL AND GAS**

2013 Oil and gas production not yet calculated regarding reserves

OIL REMAINING RESERVE (MMSTB)

TOTAL (3P) = 7,549.81

### Compared to 2012

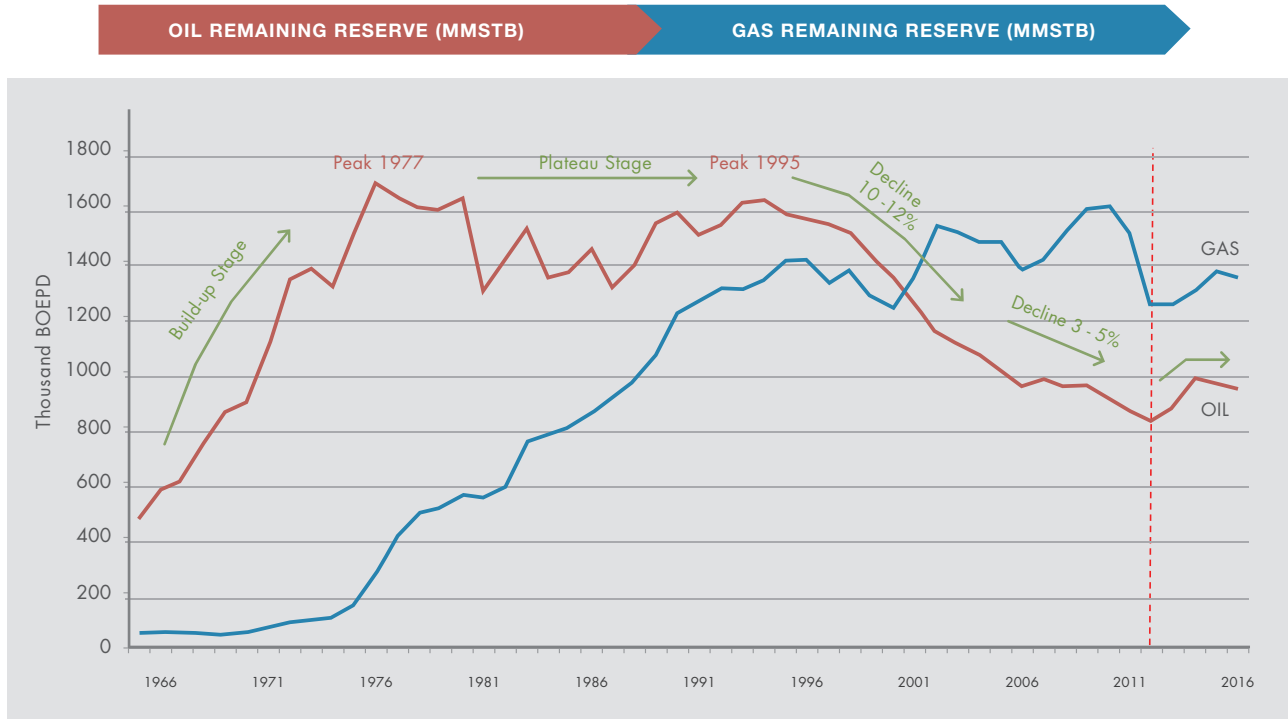
**GAS REMAINING RESERVE (MMSTB)**

TOTAL (3P) = 150.39

### Compared to 2012

16

# OIL AND GAS PRODUCTION PROFILE



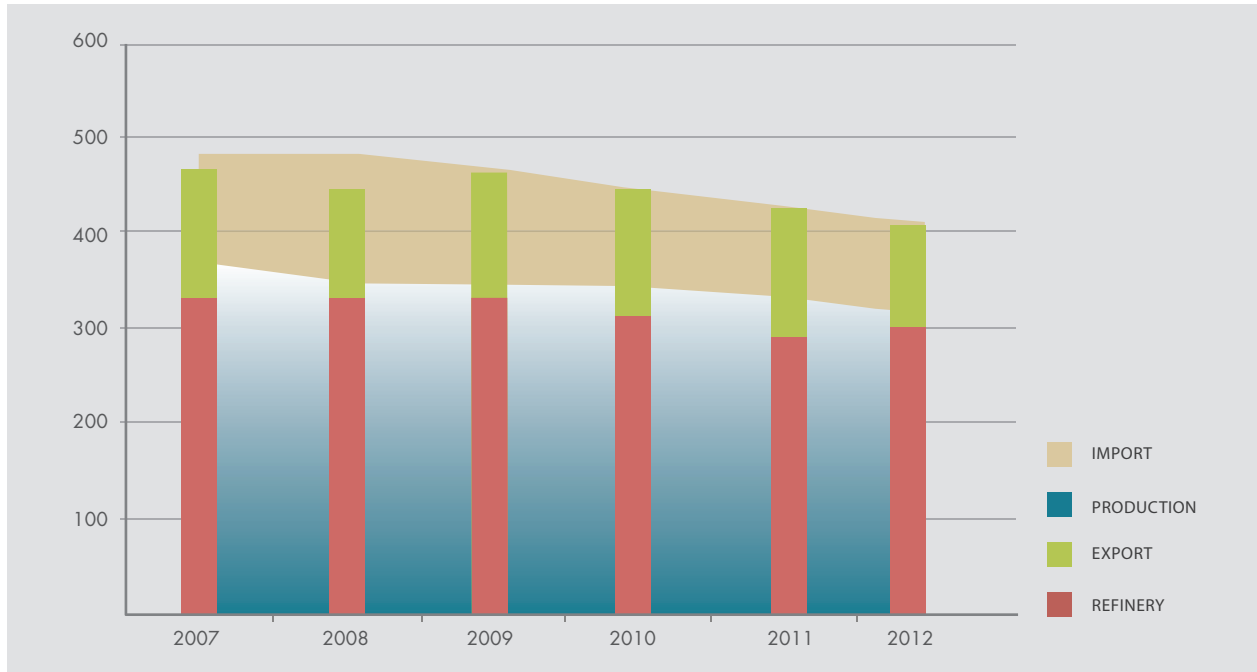
Source: SKK Migas by 2013, reprocessed by NEC

Trend of the upstream oil and gas activities

- Dominated by offshore locations
- The invention of reserves dominated by gas
- CBM start to developed
- Oil and gas exploration on the deep sea in the eastern region

# CRUDE OIL SUPPLY

(Thousand Bbl)

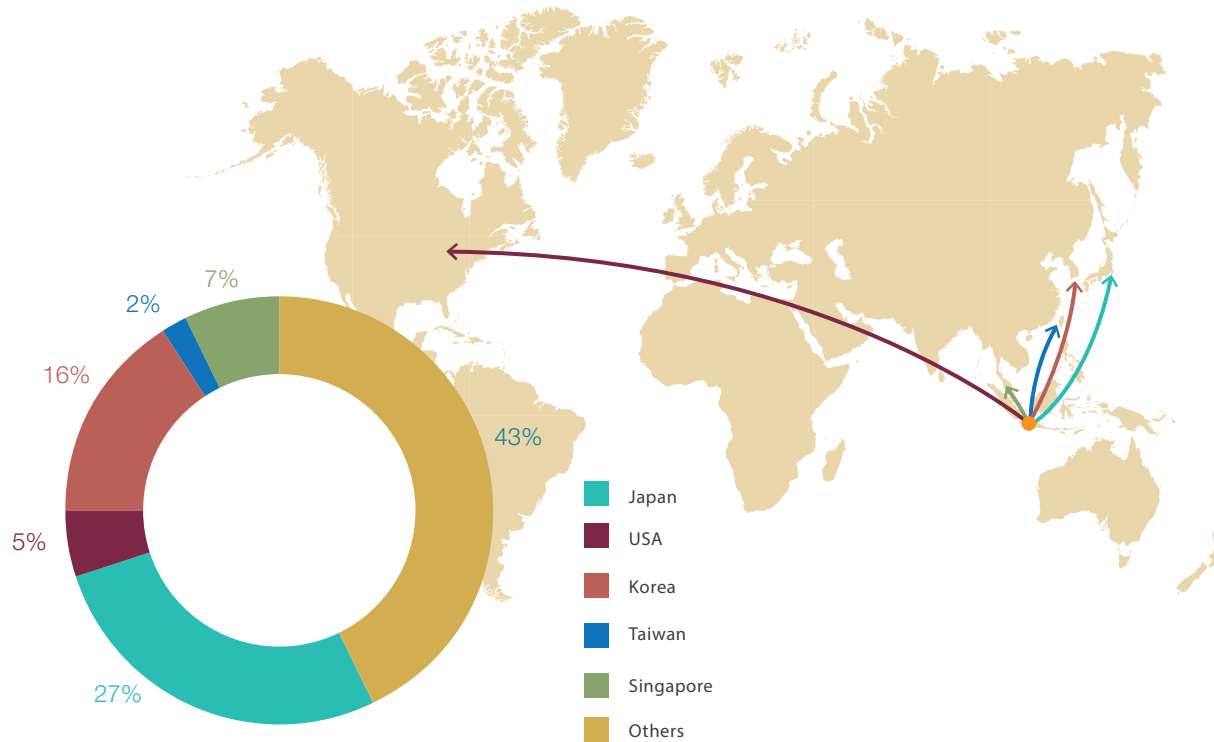


Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

- Note :
- Discrepancy between production, export, import and domestic demand due to the existing stock
  - The role of imported crude in 2012 was amounted to 32 % from the total that goes to the oil refinery



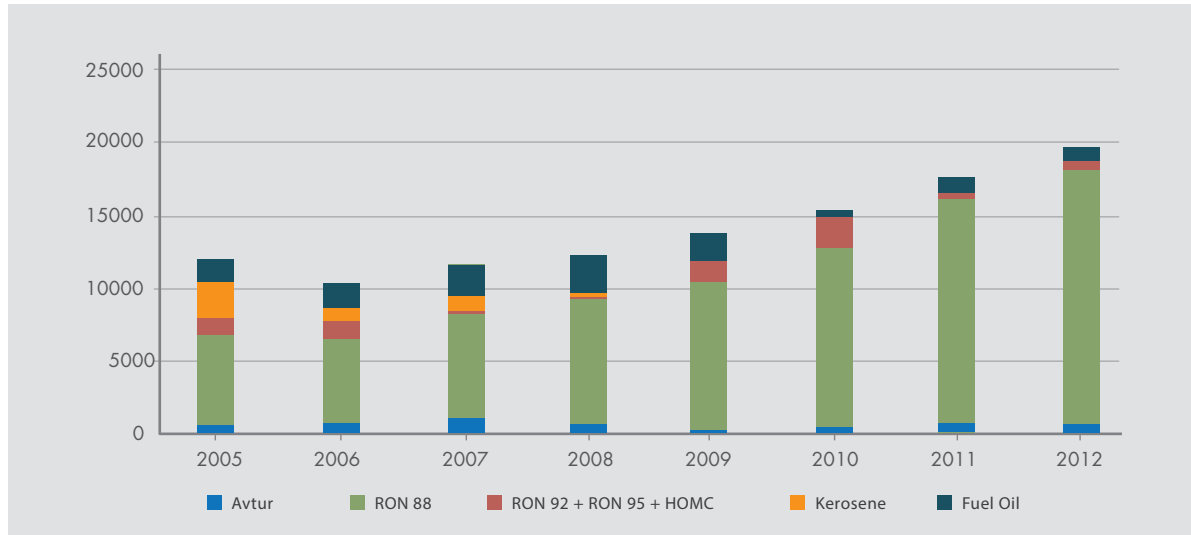
# OIL AND GAS PRODUCTION PROFILE



Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

# FUEL IMPORTS

(Thousand KL)



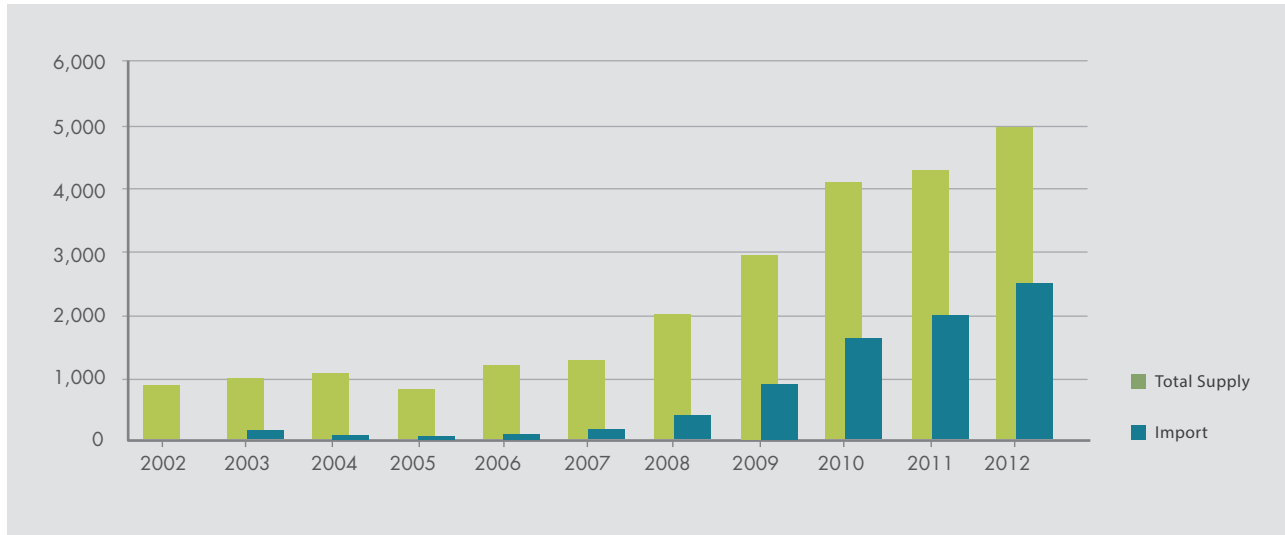
Source: Ministry of Energy and Mineral Resources 2012, reprocessed by NEC

## Note :

- Share of fuel imports in 2012 amounted to 54.25% (179.98 MBOE) of the total domestic fuel oil consumption
- The value of imports of RON 88 (premium) increased along with the increase in domestic demand
- Kerosene imports declined and abolished in 2009 due to LPG conversion policy

# LPG IMPORTS

(Kilo Ton)



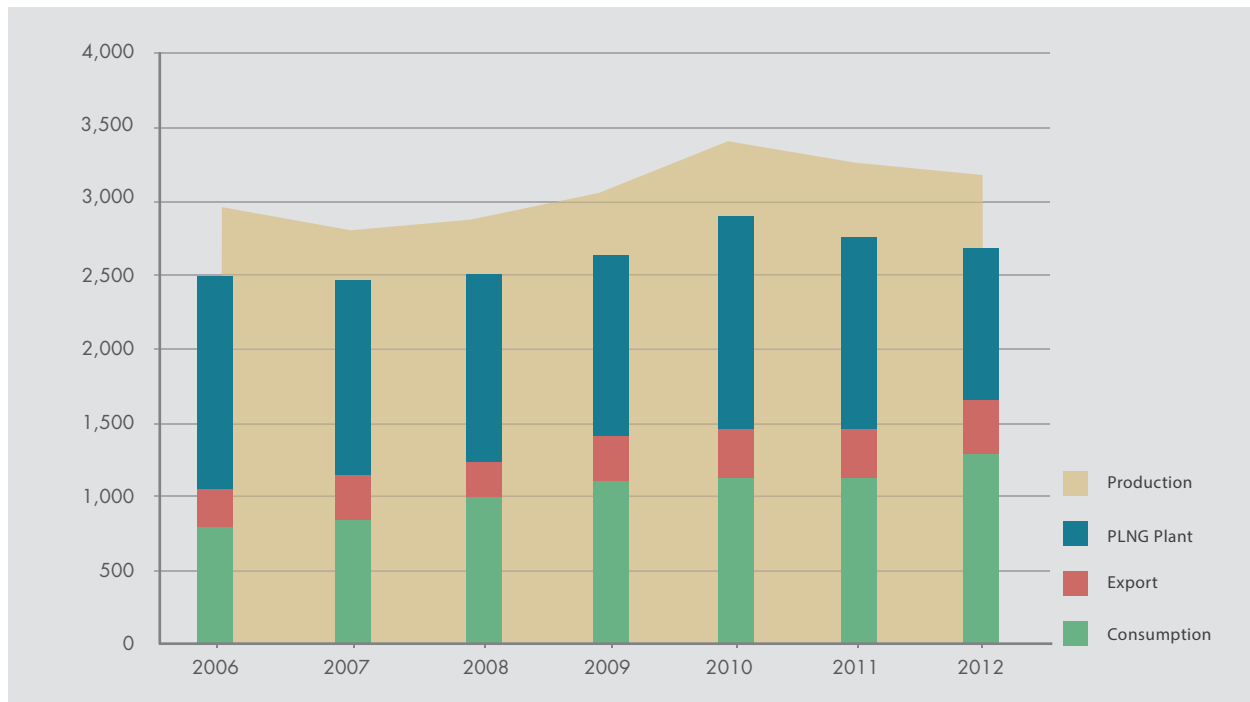
Source: Ministry of Energy and Mineral Resources 2012, reprocessed by NEC

Note :

- Share of LPG imports in 2012 amounted to 49.73% (21.3 MBOE) of the total LPG supply
- The value of LPG supply increased along with the increasing the demand in residential sector due to LPG conversion policy

# GAS SUPPLY

(Thousand MMSCF)



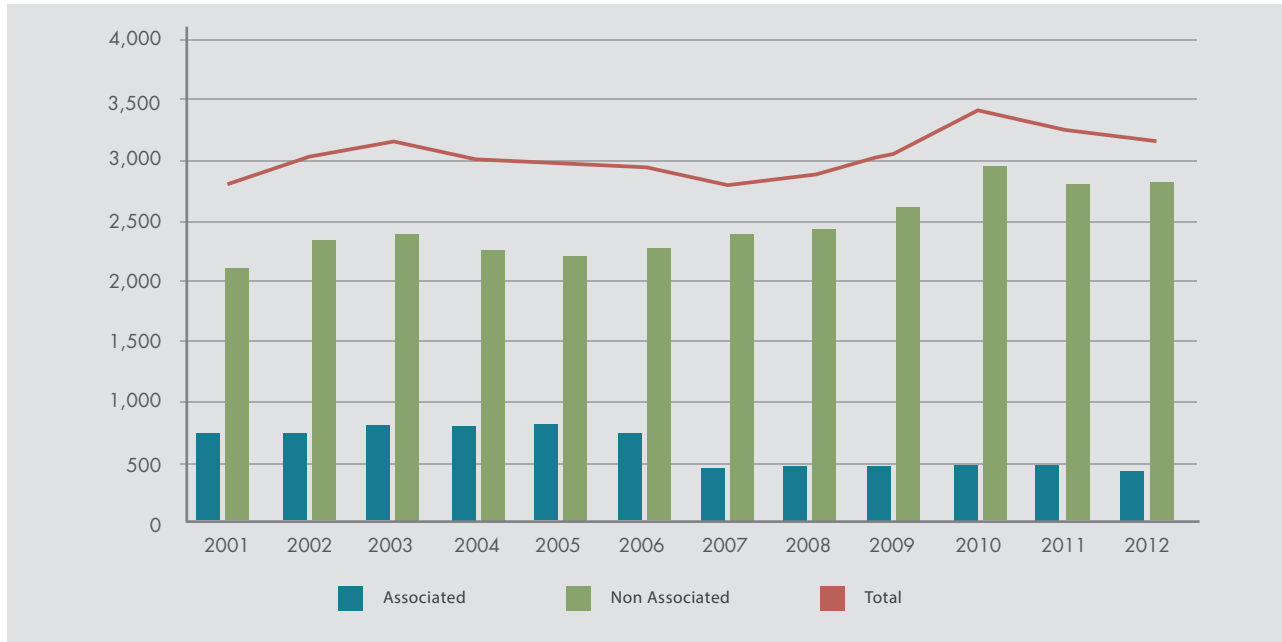
Source: Ministry of Energy and Mineral Resources 2012, reprocessed by NEC

Note:

- Export means export of gas through pipeline
- Products of LNG refinery allocated for exports
- Discrepancy between the production and the total usage of gas due to gas reinjection, own use and flare

# GAS PRODUCTION

(Thousand MMSCF)



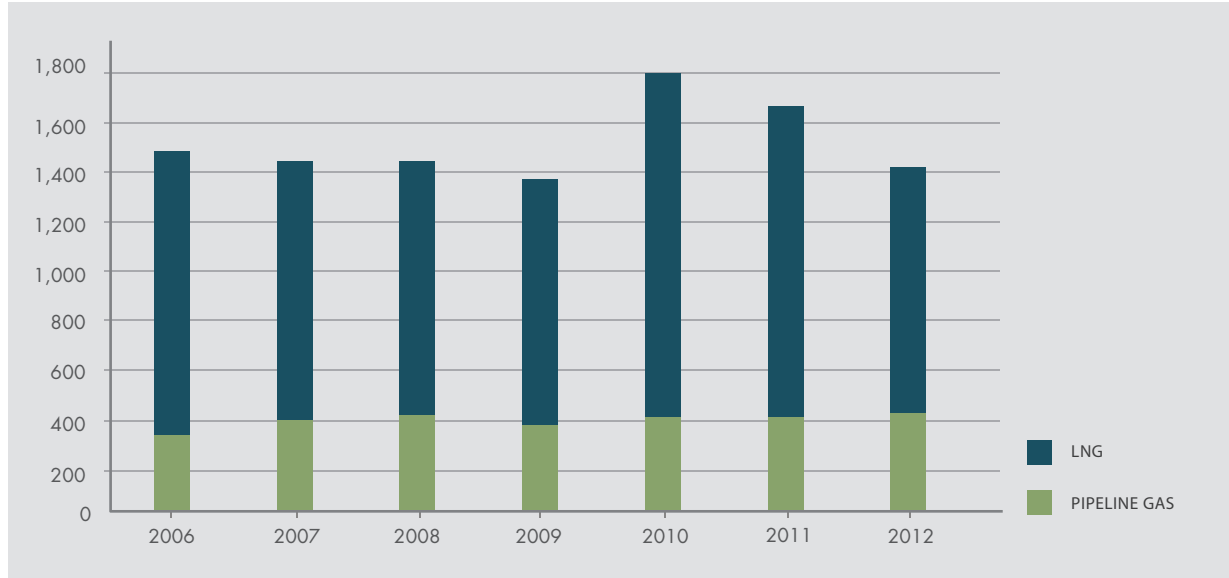
Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

Note :

Associated defined as a natural gas found in association with oil, either dissolved in the oil or as a cap of free gas above the oil.

# NATURAL GAS EXPORT

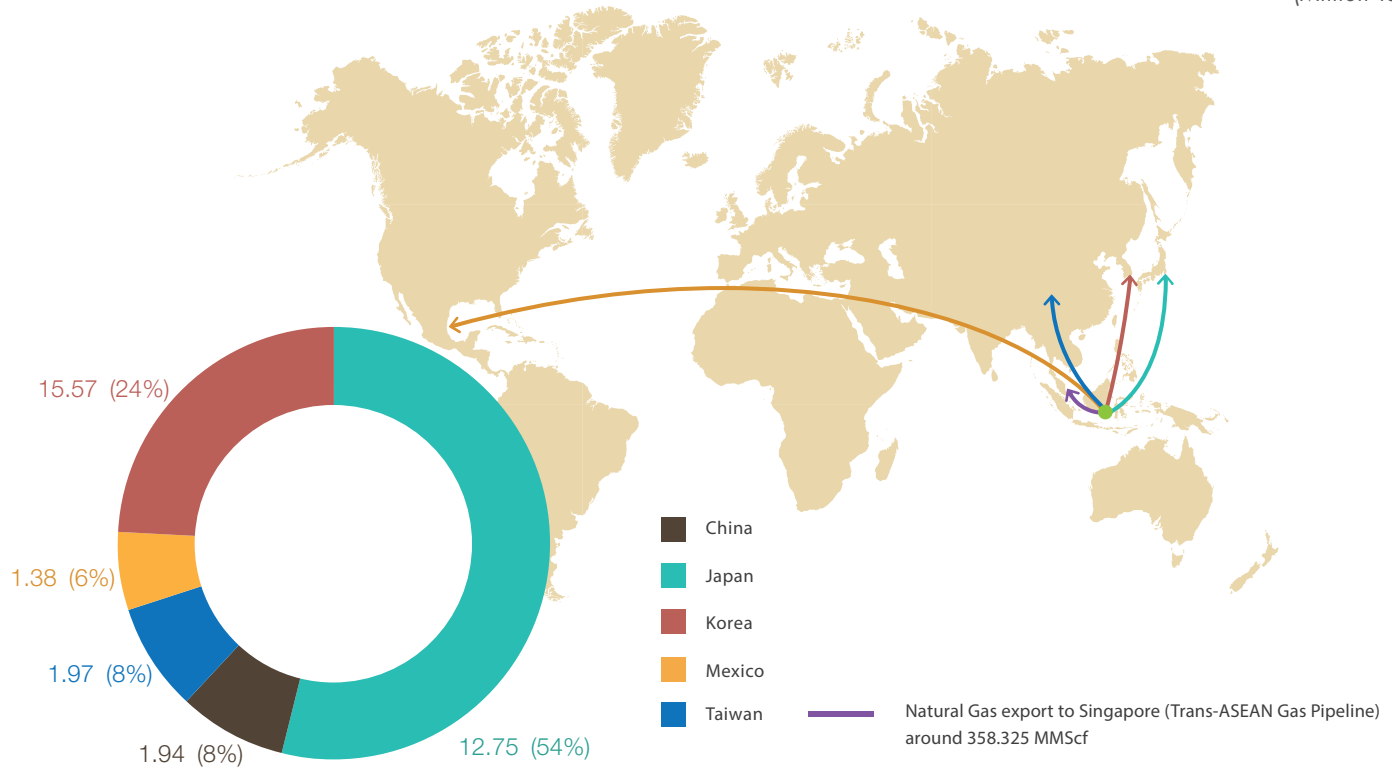
(Thousand MMSCF)



Source: Ministry of Energy and Mineral Resources 2012, reprocessed by NEC

# GAS AND LNG EXPORTS

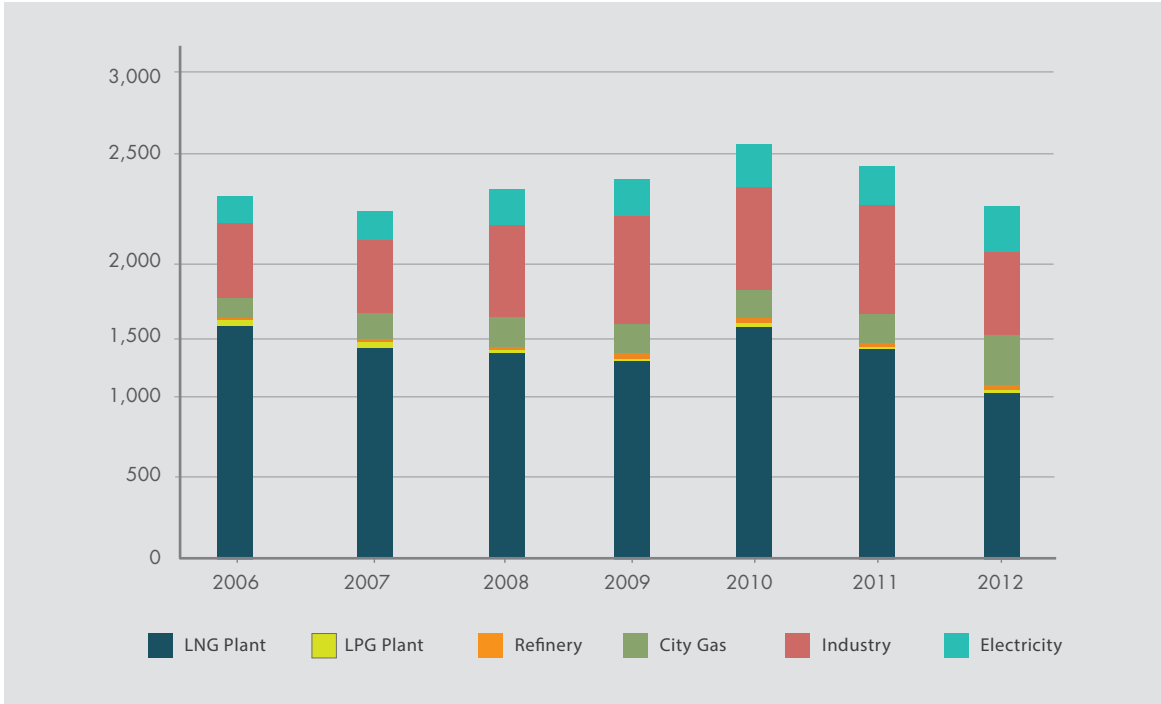
(Million Ton)



Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

# NATURAL GAS CONSUMPTION

(Thousand MMSCF)

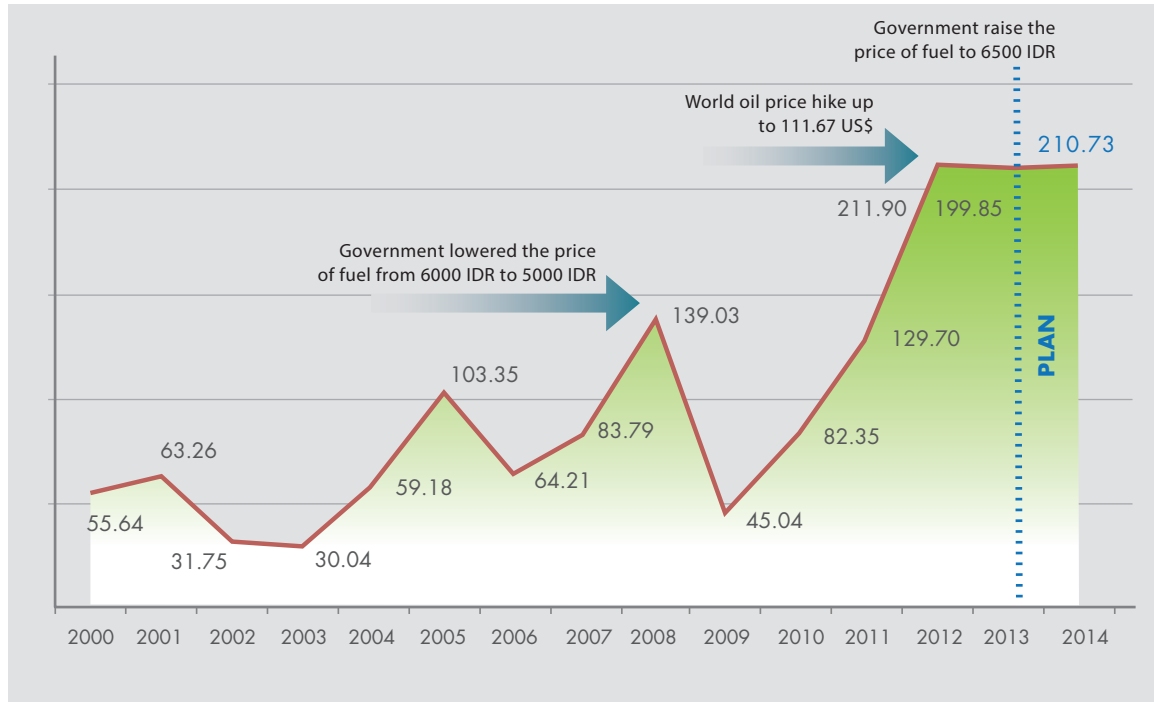


Source: Ministry of Energy and Mineral Resources 2012, reprocessed by NEC



# FUEL OIL SUBSIDY

(Trillion IDR)

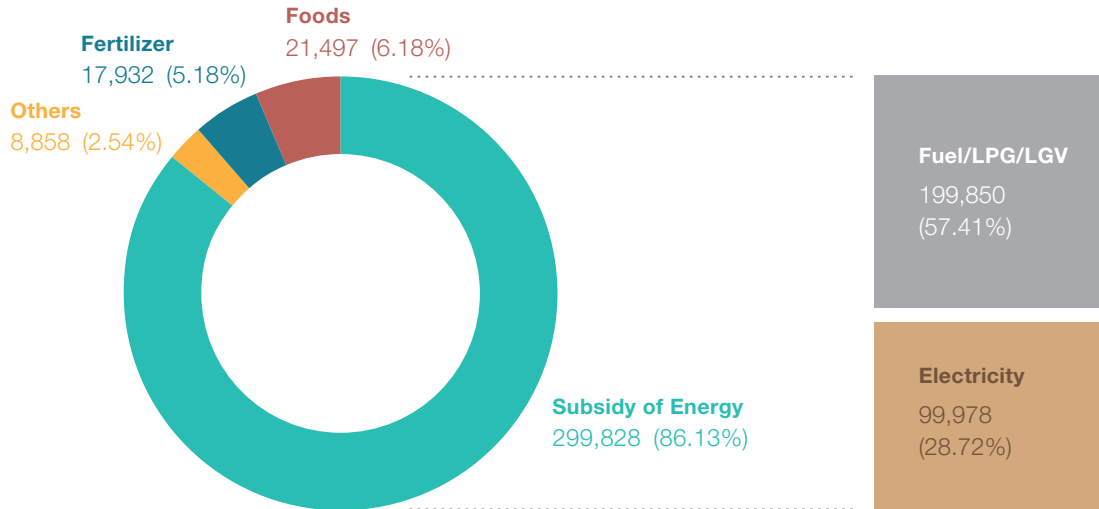


Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

Note: Rise in oil prices due to geopolitical crisis in the Middle East and Libya

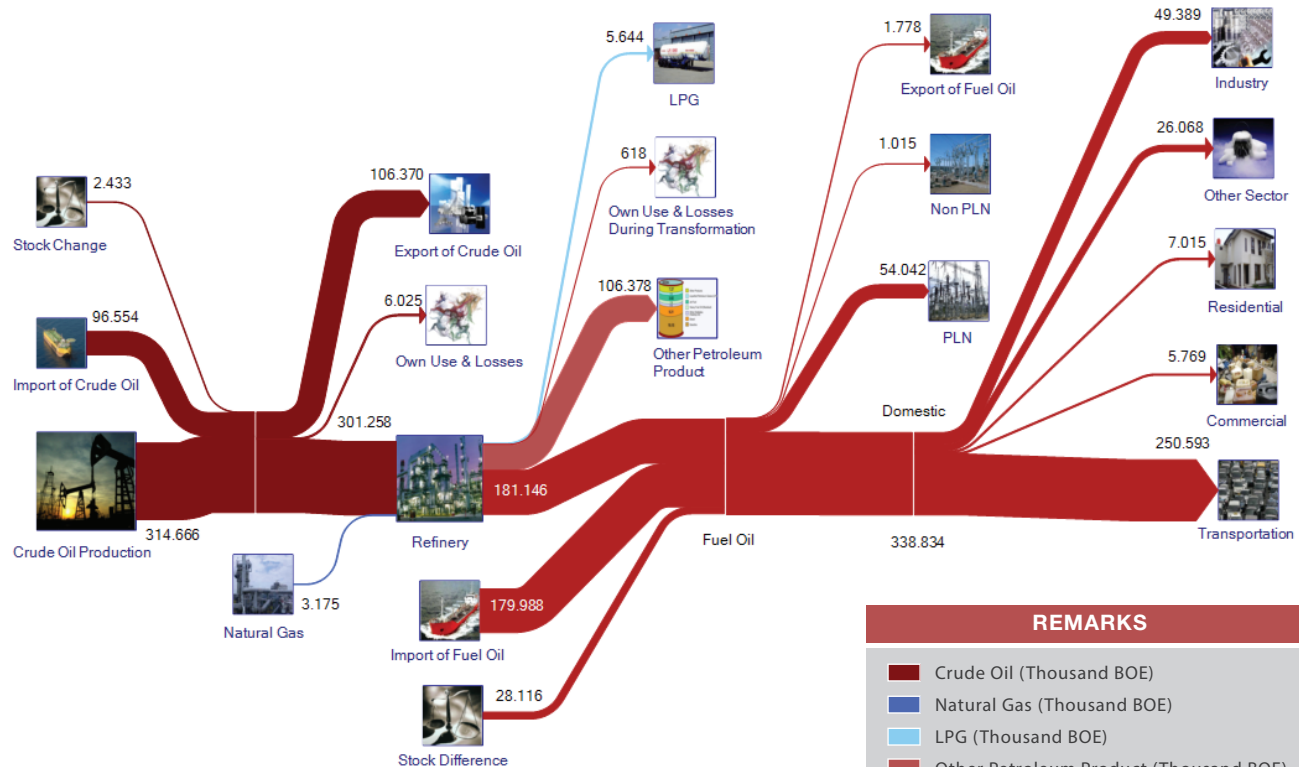
# SHARES OF TOTAL ENERGY SUBSIDY, 2013

(Trillion Rupiah)



Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

# OIL FLOW, 2013\*)



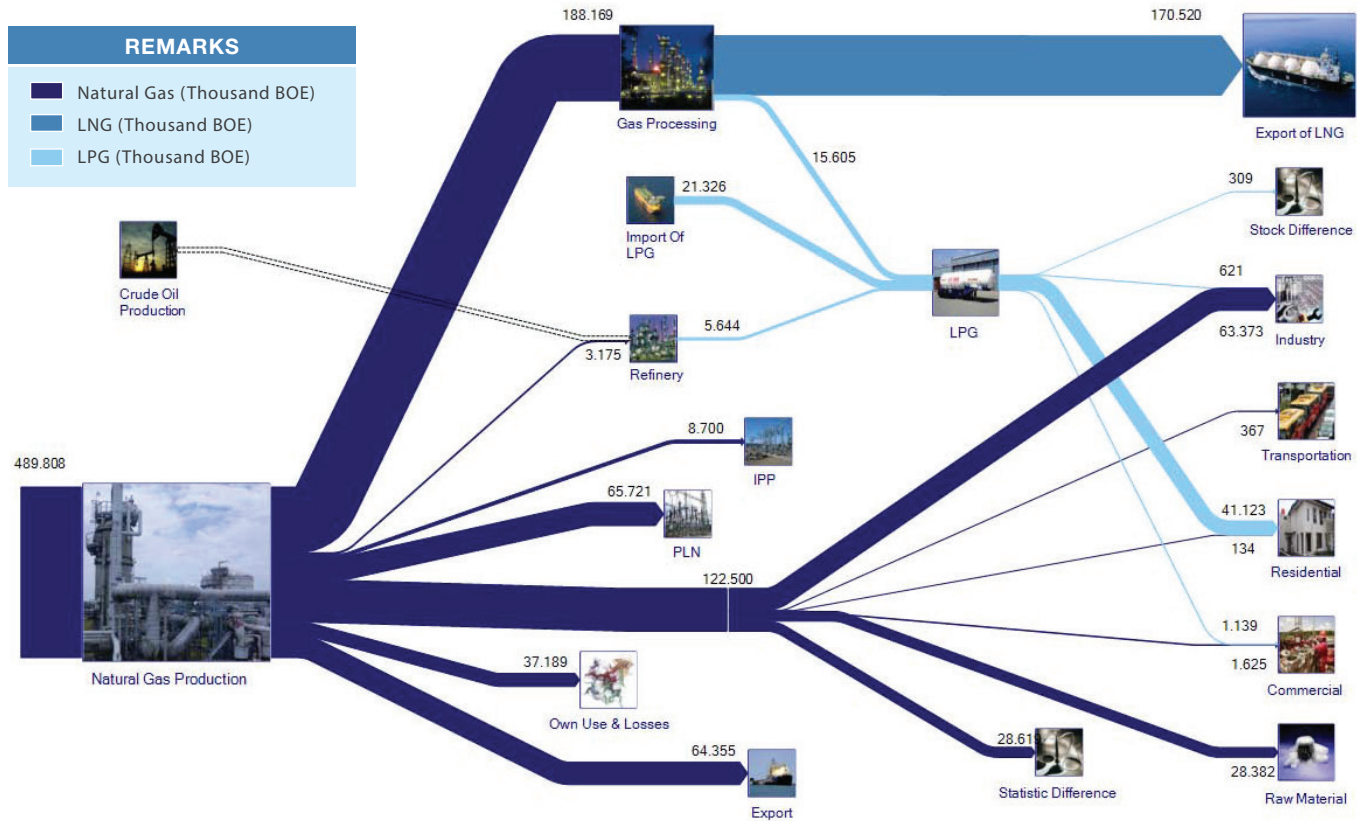
## REMARKS

- Crude Oil (Thousand BOE)
- Natural Gas (Thousand BOE)
- LPG (Thousand BOE)
- Other Petroleum Product (Thousand BOE)
- Fuel Oil (Thousand BOE)

Source: National Energy Council

\*) in the process of updating data

# GAS FLOW, 2013\*)



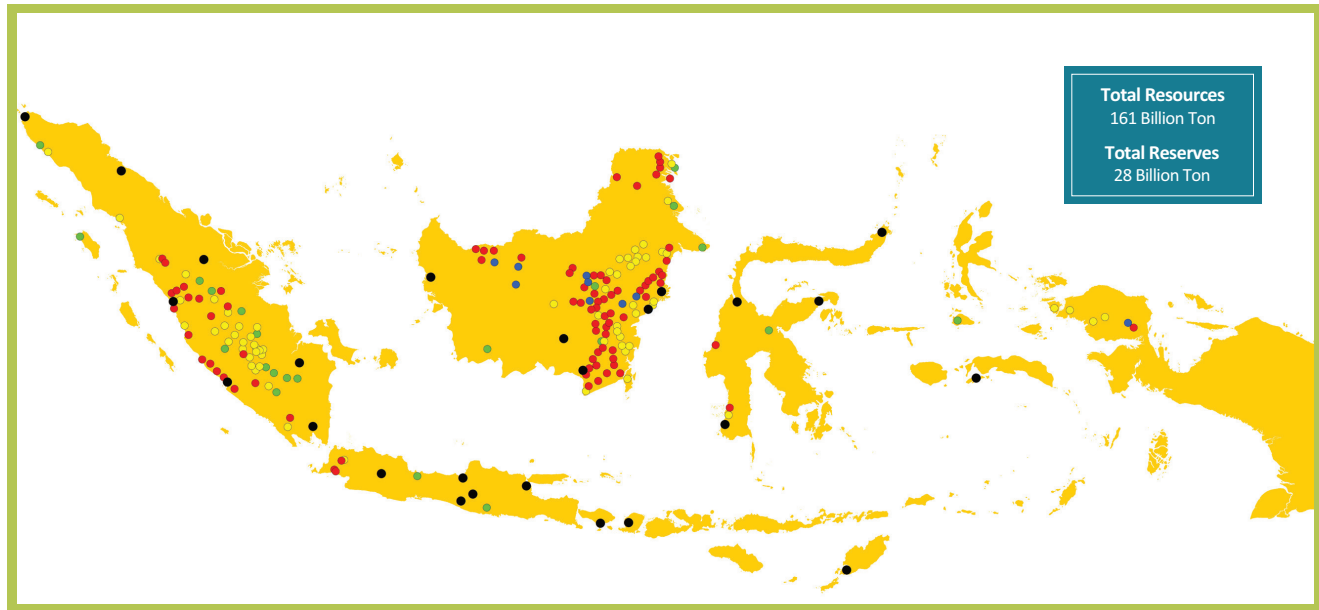
Source: National Energy Council

\*) in the process of updating data



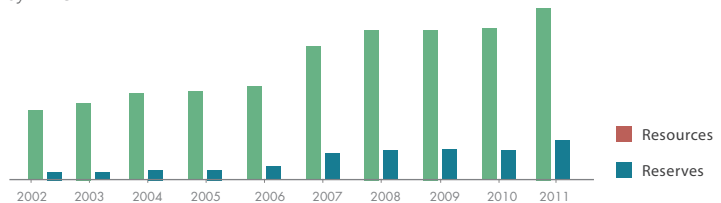
**COAL**

# COAL RESOURCES AND RESERVES



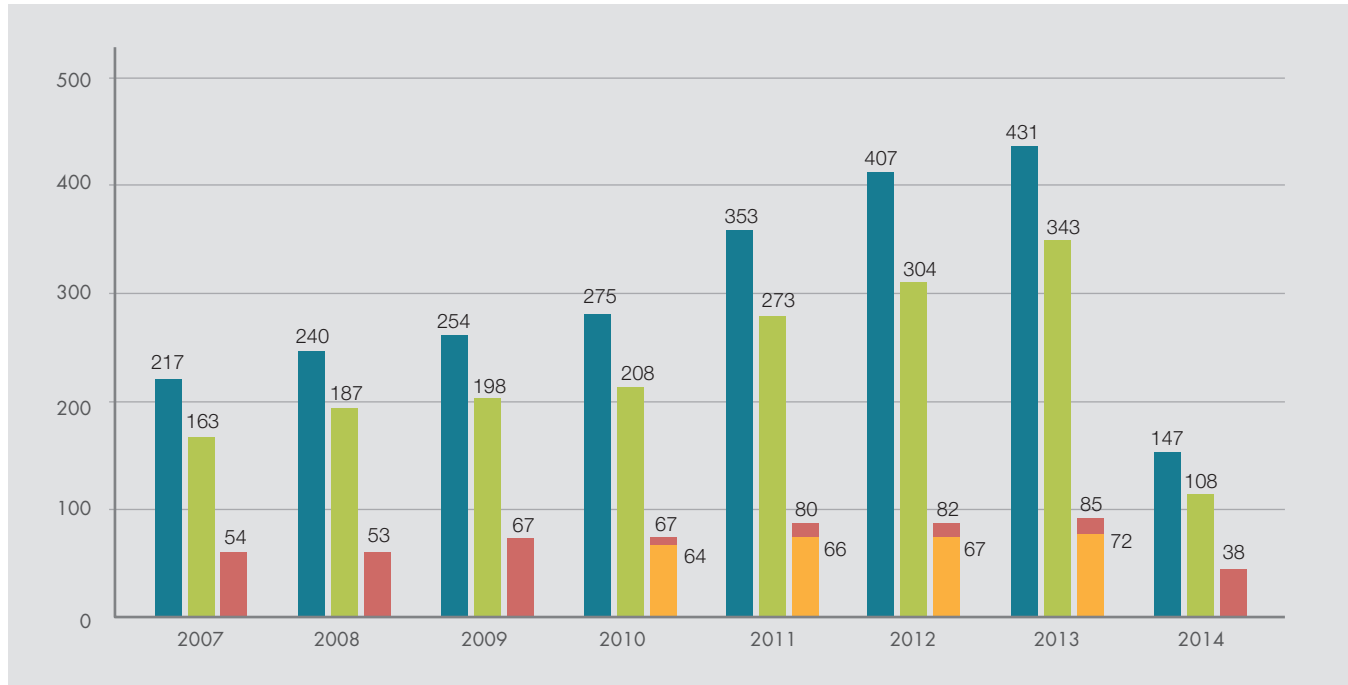
Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

- Very High (> 7,100 cal/gr)
- High (6,100 - 7,100 cal/gr)
- Medium (5,100 - 6,100 cal/gr)
- Low (< 5,100 cal/gr)
- Reserves



# COAL UTILIZATION

(Million Ton)



Source: Ministry of Energy and Mineral Resources 2012, reprocessed by NEC

\*) April 2014

■ Production

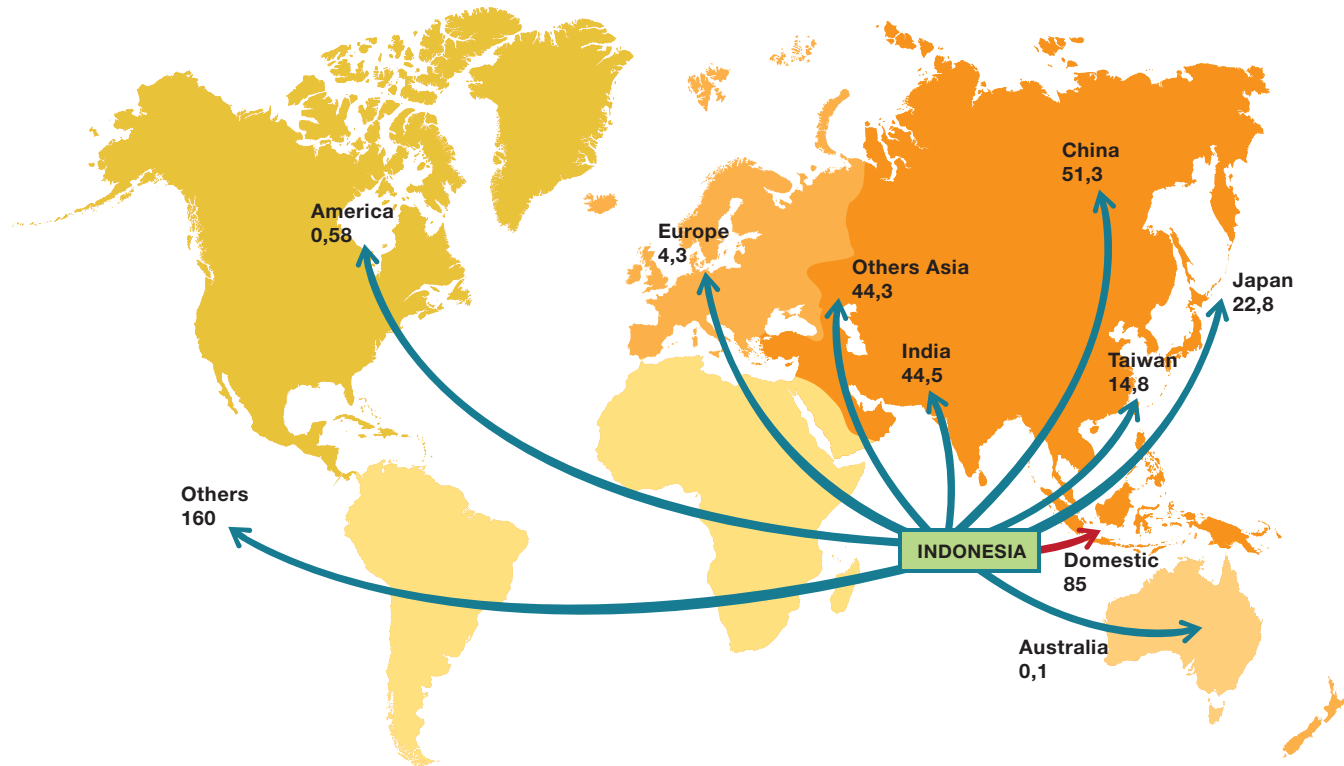
■ Export

■ Domestic

■ DMO

# COAL EXPORT BY COUNTRY DESTINATION

(Million Ton)

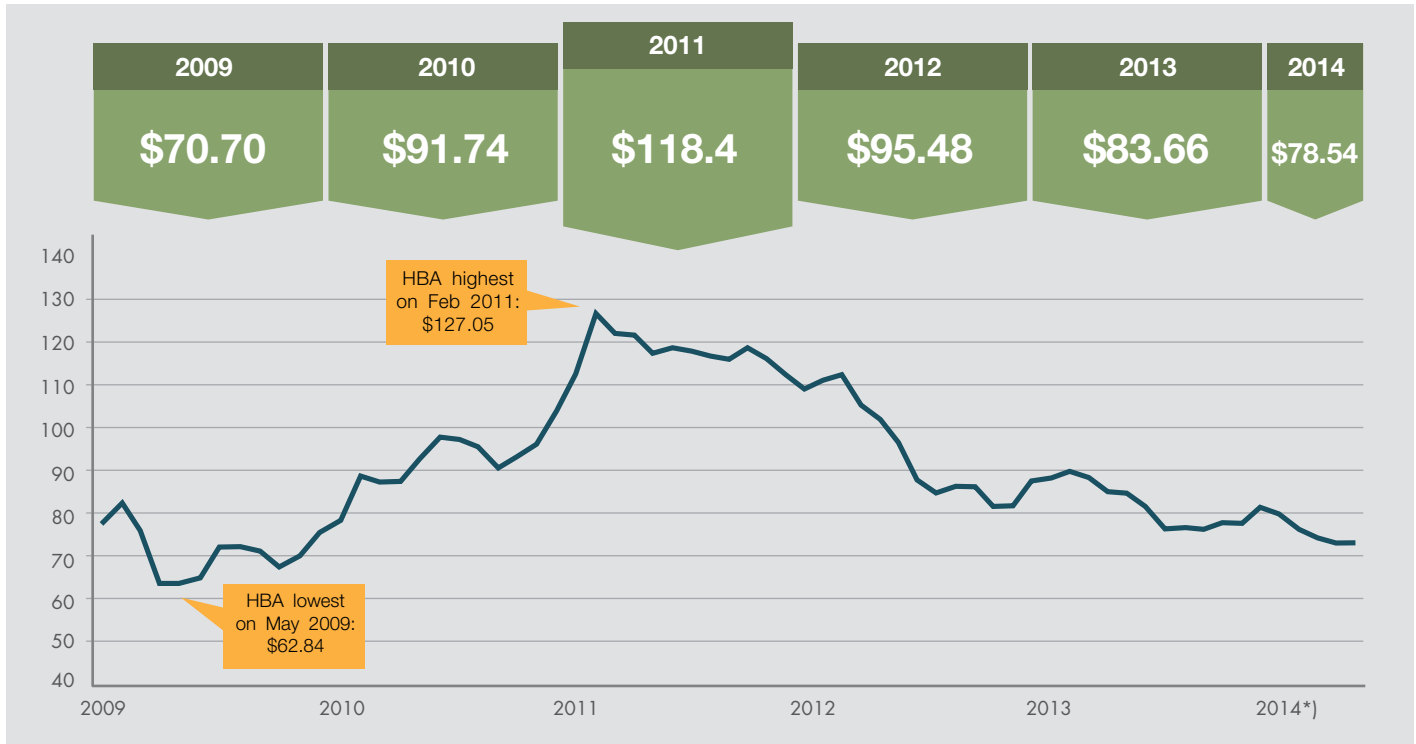


Source: Ministry of Energy and Mineral Resources, reprocessed by NEC



# COAL REFERENCE PRICE (HBA)

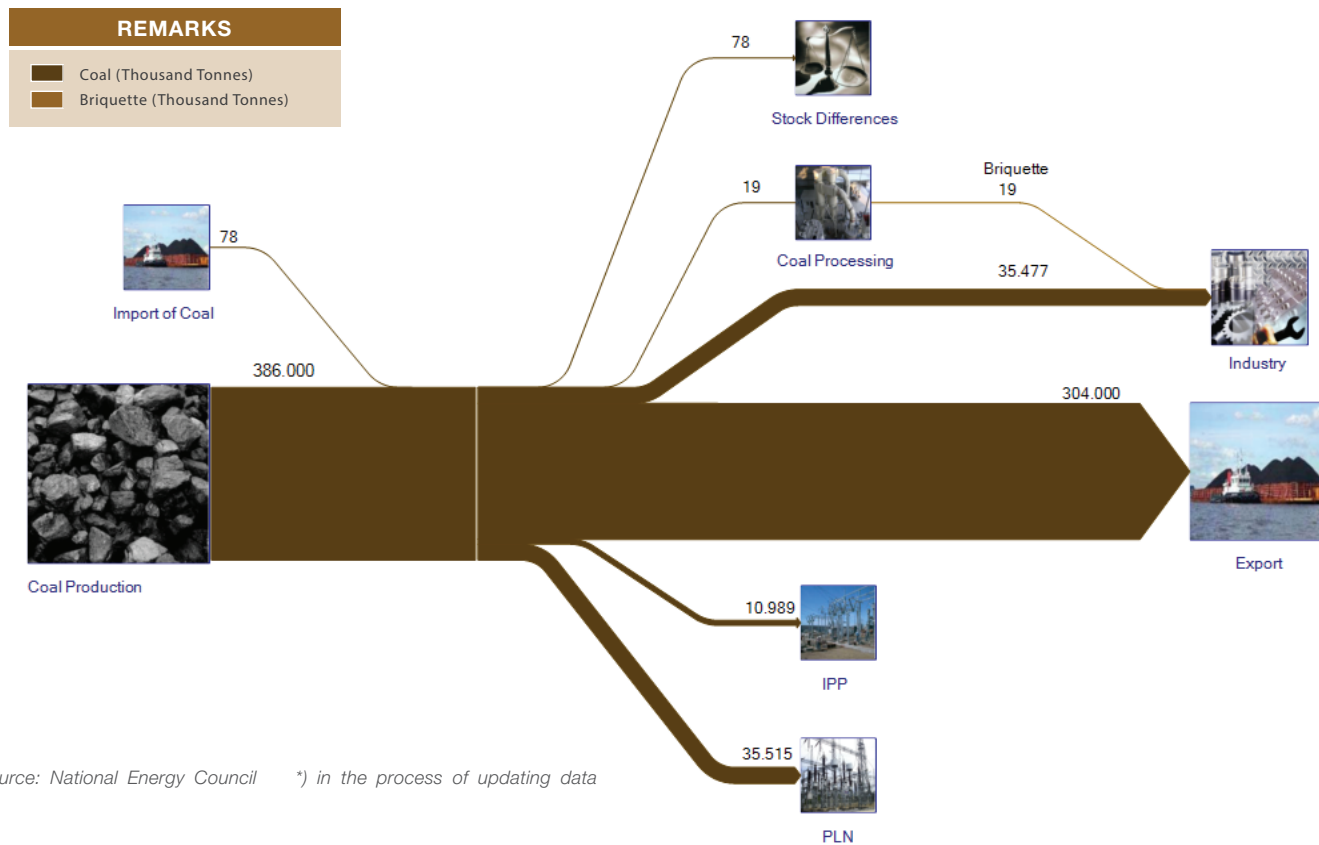
(US Dollar)



Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

\*) Juni 2014

# COAL FLOW, 2013<sup>\*)</sup>



A solid orange horizontal bar spanning the width of the page.

**ELECTRICITY**

# ELECTRICITY

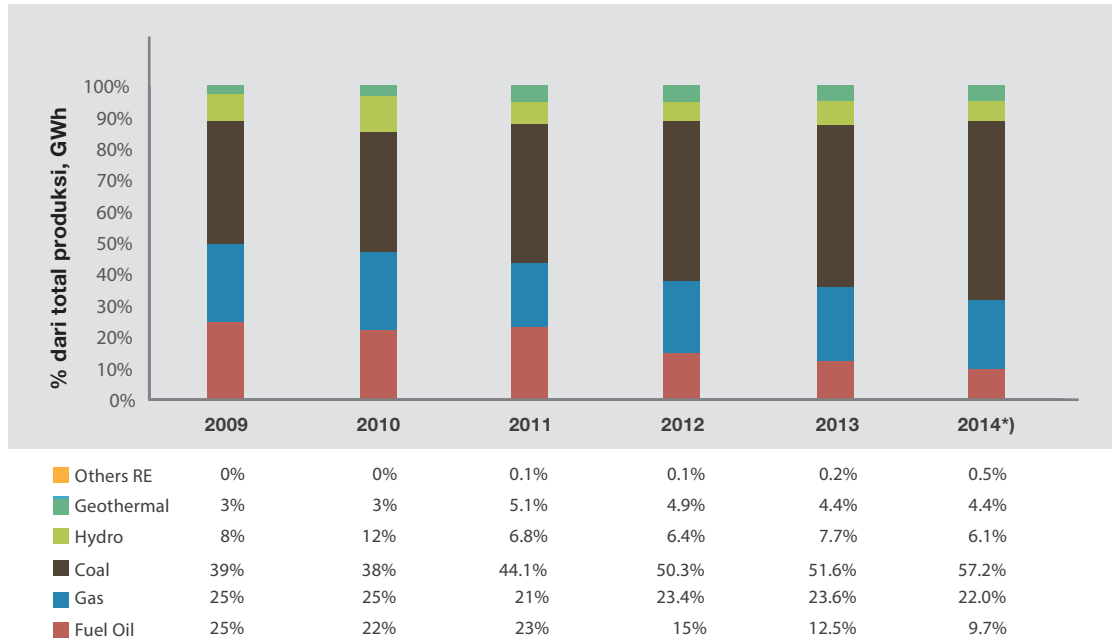
| DESCRIPTION                                            | UNIT              | 2011     | 2012     | 2013    |
|--------------------------------------------------------|-------------------|----------|----------|---------|
| Demand Growth                                          | %                 | 10.1     | 8.4      | 8.6     |
| Electrification Ratio                                  | %                 | 72.9     | 76.6     | 80.5    |
| Village Electrification Ratio                          | %                 | 96       | 96.7     | 97.8    |
| Total Installed Capacity                               | MW                | 39,885   | 44,124   | 48,161  |
| a. PLN                                                 | MW                | 30,529   | 32,108   | 35,564  |
| b. IPP                                                 | MW                | 7,653    | 10,287   | 10,718  |
| c. <i>Private Power Utilities</i> (PPU)                | MW                | 1,704    | 1,729    | 1,729   |
| PLN Electricity Production and Purchase of Electricity | GWh               | 175,213  | 193,663  | 207,409 |
| Rural Electricity                                      |                   |          |          |         |
| a. Sub Station Distribution                            | MVA               | 368      | 249      | 216.8   |
| b. Distribution Network                                | KMs               | 17,570.7 | 11,311.5 | 9,244.3 |
| c. Cheap and Efficient Electricity                     | RTS <sup>*)</sup> | -        | 60,702   | 95,227  |

Source : Ministry of Energy and Mineral Resources & National Electric Company (PLN)

\*) RTS = Target Household (<450 Watt)

# POWER PLANT ENERGY MIX

(%)



Source : Ministry of Energy and Mineral Resources, reprocessed by NEC

\*) Target 2014

# POWER PLANTS DEVELOPMENT PLANNING



Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

- |                                                           |                                                                        |                                                                 |                                                                 |
|-----------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------|
| <span style="color: lightblue;">●</span> Hydropower Plant | <span style="color: darkgreen;">●</span> Coal Gasification Power Plant | <span style="color: darkpurple;">●</span> Minihydro Power Plant | <span style="color: yellow;">●</span> Steam Power Plant         |
| <span style="color: red;">●</span> Gas Power Plant        | <span style="color: lightgreen;">●</span> Steam Gas Power Plant        | <span style="color: pink;">●</span> Geothermal Power Plant      | <span style="color: darkblue;">●</span> Micro Hydro Power Plant |

# ELECTRICITY TRANSMISSION NETWORK

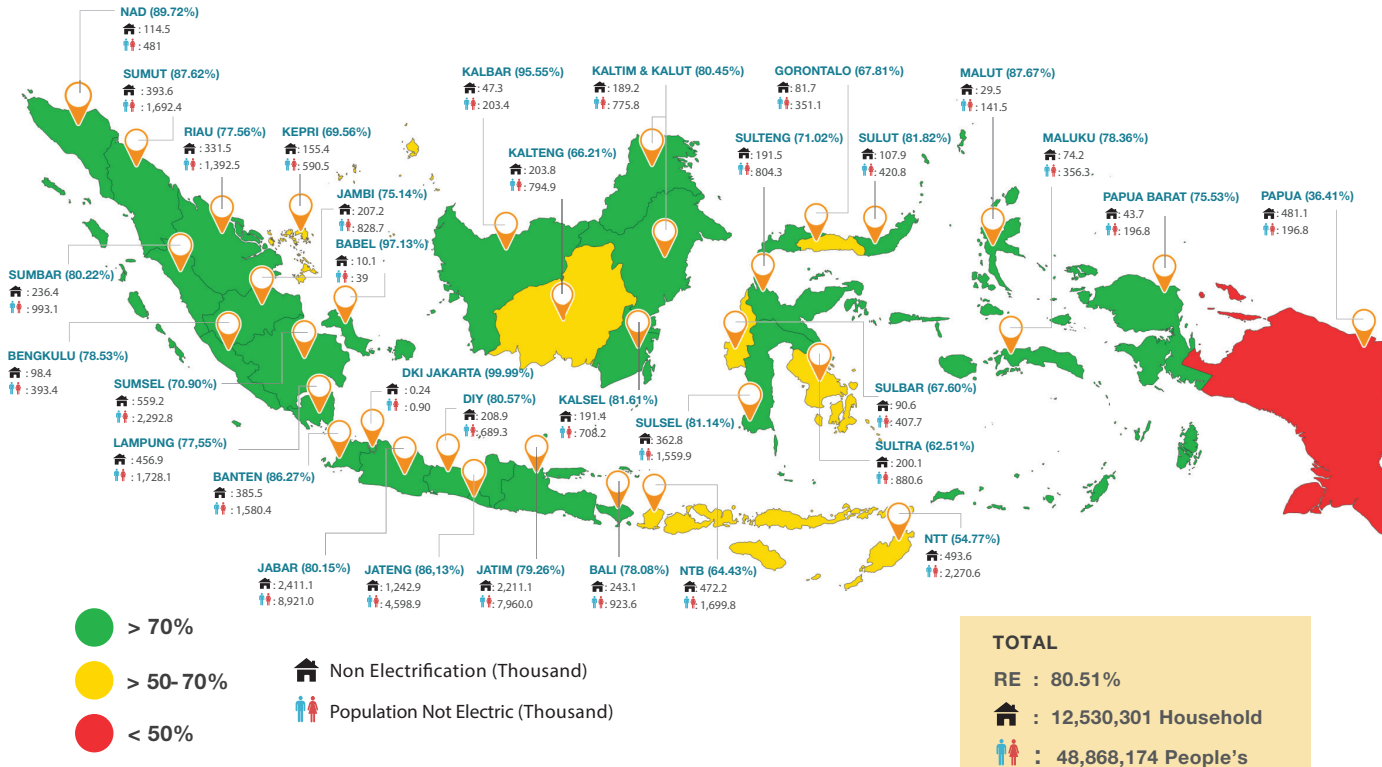


Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

Existing Transmission Line

Transmission Line Plan

# ELECTRIFICATION RATIO



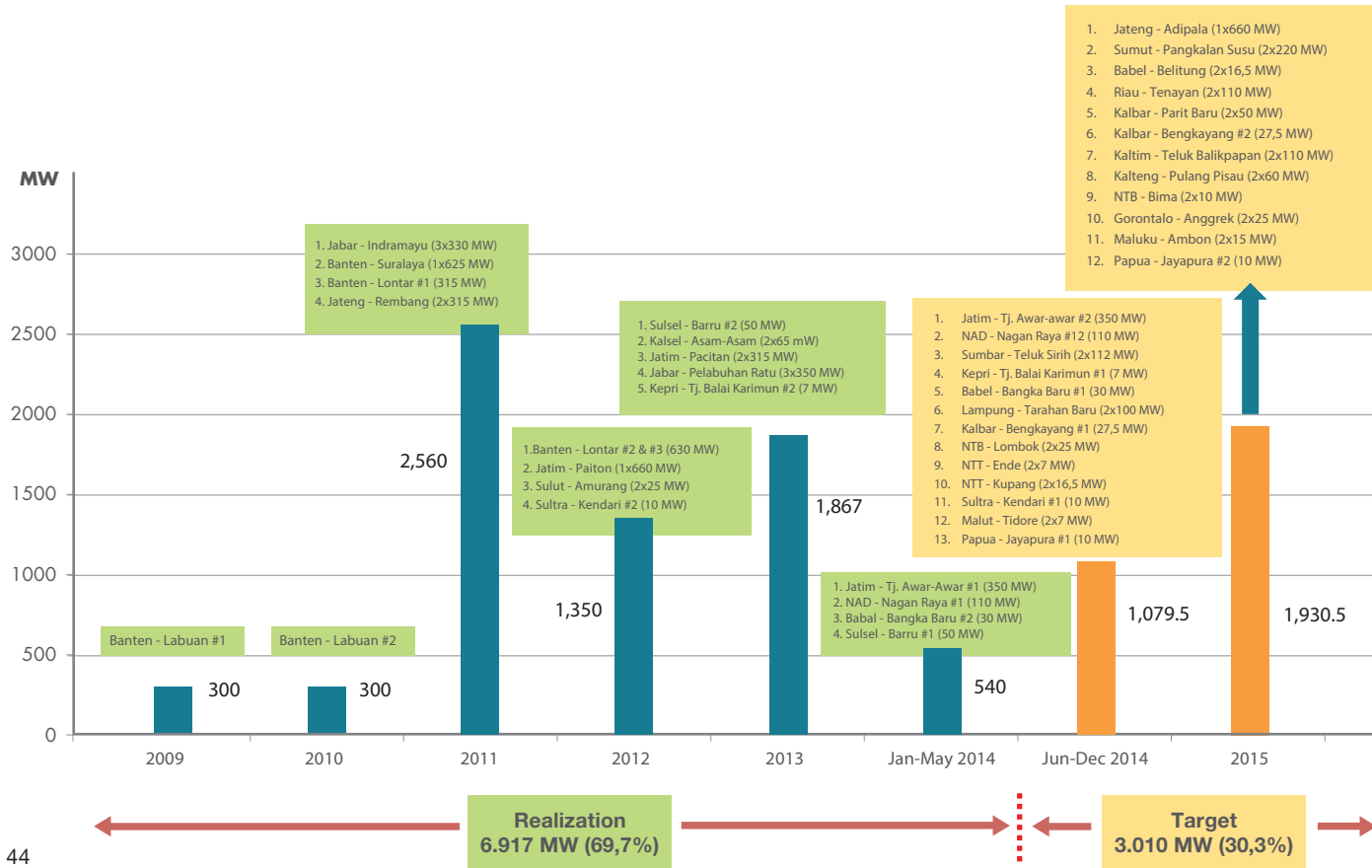
Source: Ministry of Energy and Mineral Resources and BPS, reprocessed by NEC



A solid orange horizontal bar spanning the width of the slide, positioned above the title text.

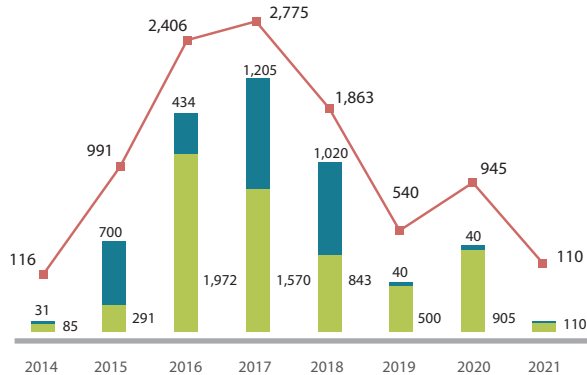
## **PROGRESS OF FTP-I AND FTP-II**

# REALIZATION AND COD TARGET OF POWER PLANT 10.000 MW FAST TRACK PROGRAM PHASE I

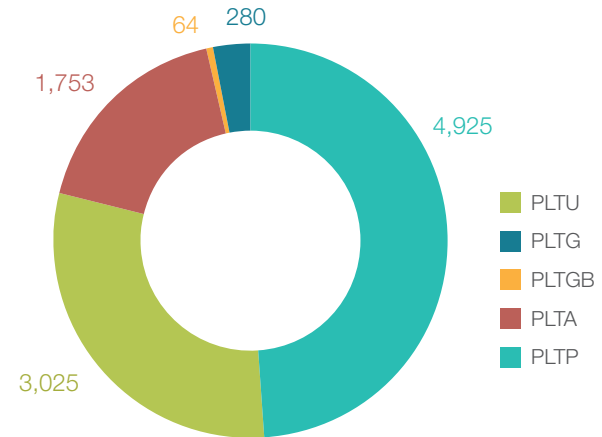


# 10,000 MW FAST TRACK PROGRAM PHASE II (FTP II)

COMMERCIAL OPERATION PLAN (MW)

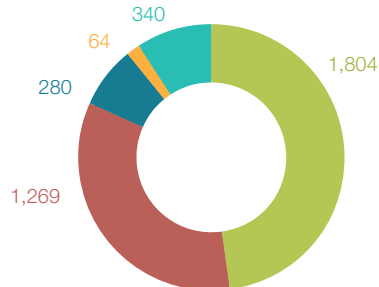


BY TYPE OF POWERPLANT (MW)

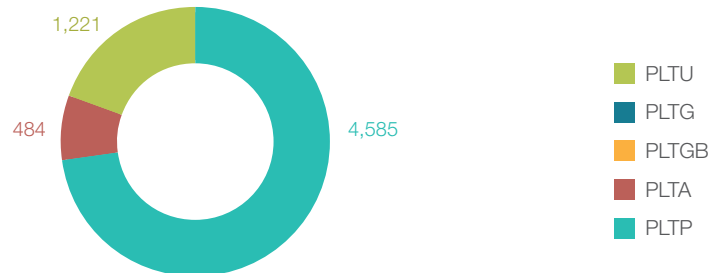


KOMPOSISI PER JENIS PEMBANGKIT

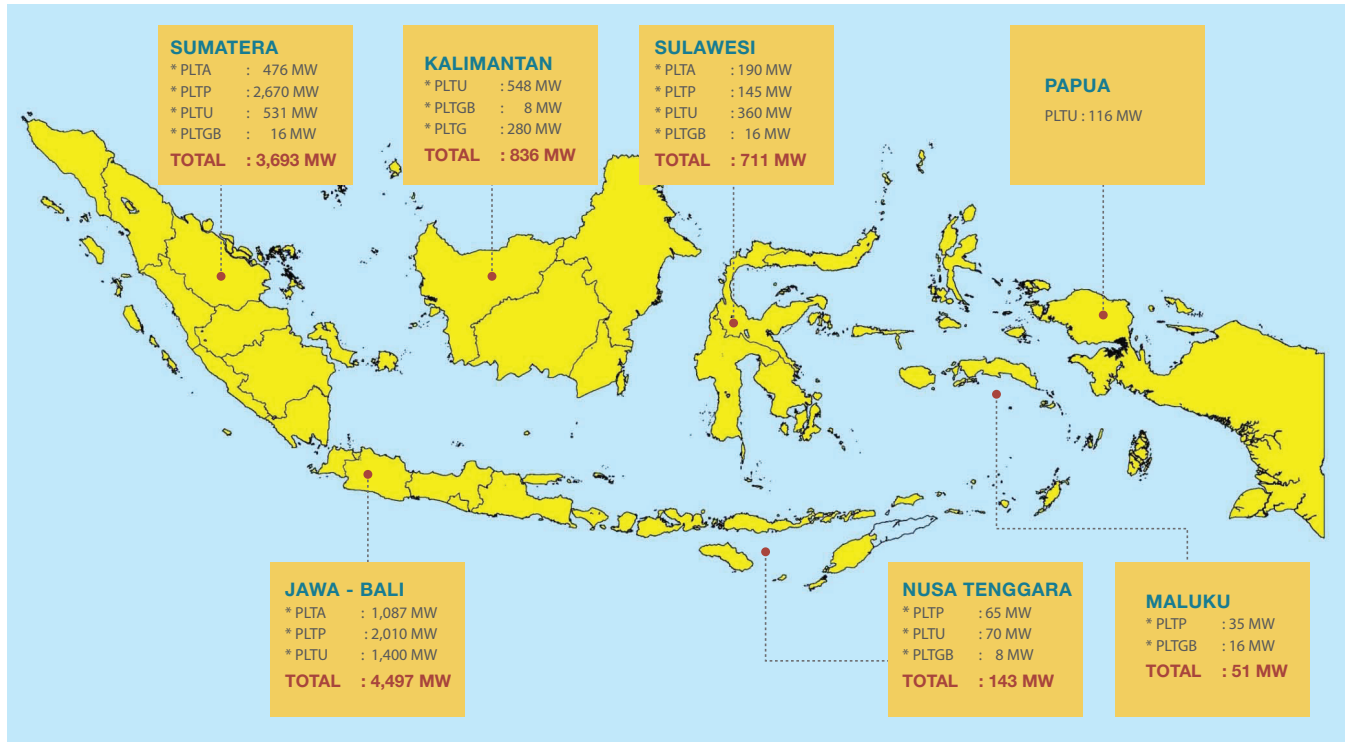
PLN (26 PROJECTS) : 3.757 MW (37%)



IPP (72 PROJECTS) : 6.290 MW (63%)



# DISTRIBUTION OF 10,000 MW PROGRAM PHASE II



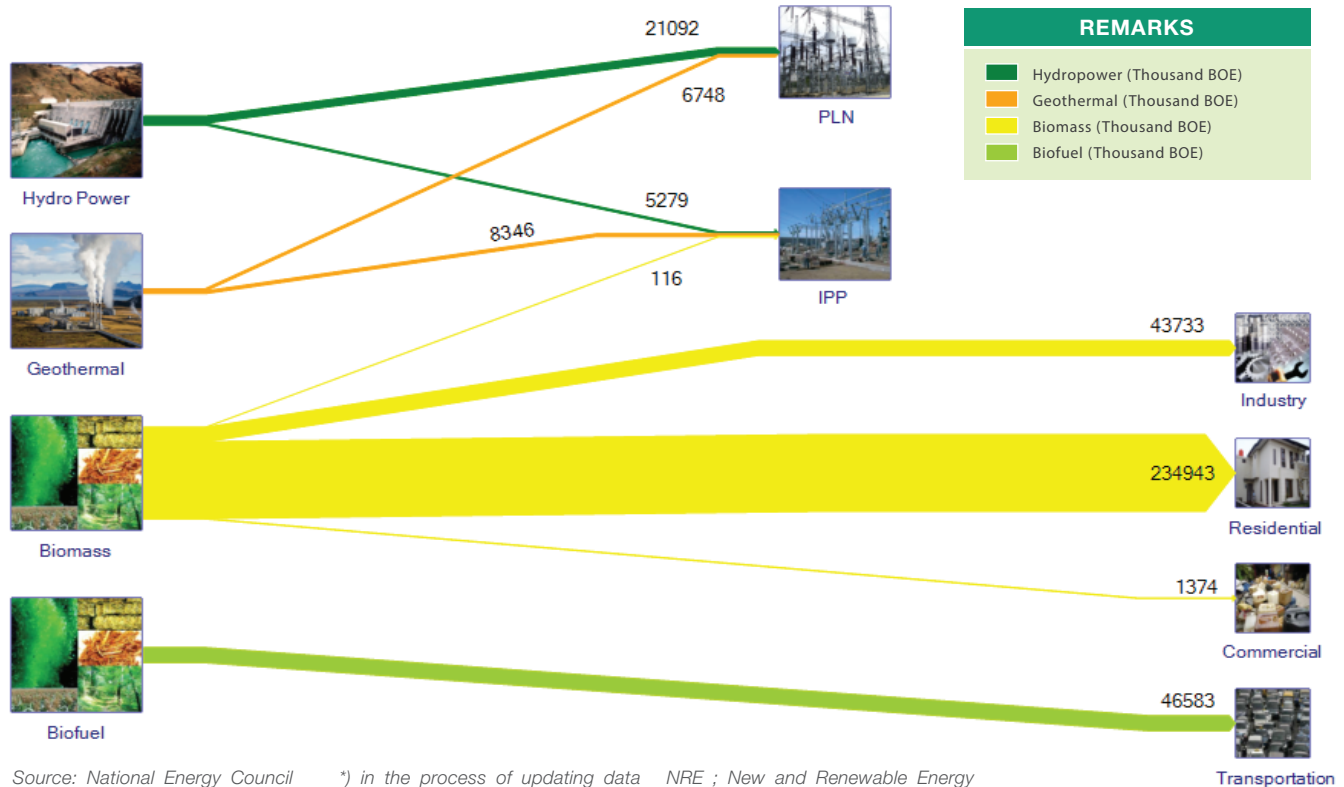
Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

\*) EMR Ministerial Regulation Number 01 year 2012

A solid orange horizontal bar spanning the width of the page, positioned above the text.

## **NEW AND RENEWABLE ENERGY**

# N R E FLOW, 2013<sup>\*)</sup>



# NEW AND RENEWABLE ENERGY RESOURCES

| NO. | TYPE            | RESOURCE                     | INSTALLED CAPACITY (MW) | RATIO (%) |
|-----|-----------------|------------------------------|-------------------------|-----------|
| 1   | 2               | 3                            | 4                       | 5 = 4/3   |
| 1   | Hydro (MW)      | 75,000MW                     | 6,848.46 MW             | 9.13%     |
| 2   | Geothermal (MW) | 29,164 MW                    | 1,341 MW                | 4.6 %     |
| 3   | Biomass (MW)    | 49,810 MW                    | 1,644.1 MW              | 3.3%      |
| 4   | Solar Energy    | 4.80 kWh/m <sup>2</sup> /day | 22.45MW                 | -         |
| 5   | Wind Energy     | 3 –6 m/s                     | 1.87 MW                 | -         |
| 6   | Ocean           | 49 GW 2)                     | 0.01 MW <sup>3)</sup>   | 0%        |
| 7   | Uranium(MW)     | 3,000 MW                     | 30 MW <sup>1)</sup>     | 0%        |

Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

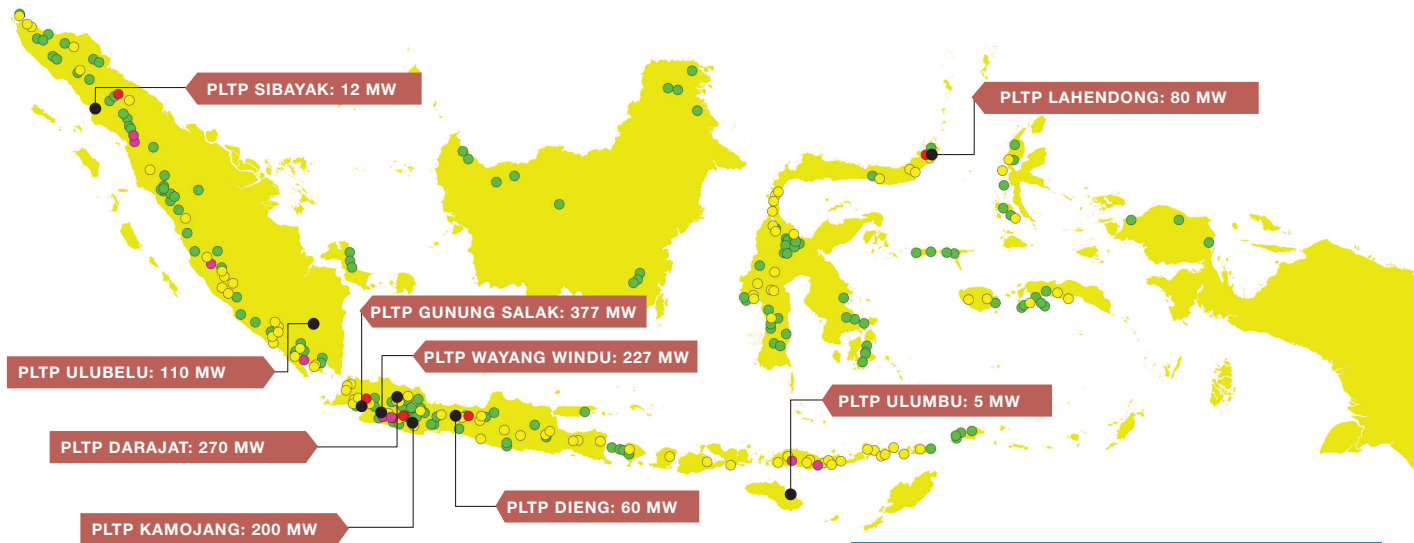
Note:

1) Research scale, non-energy

2) Source: National Energy Council

3) Research Scale: BPPT

# GEOTHERMAL POTENTIAL



Source: Geological Agency, Ministry of Energy and Mineral Resources (2012)

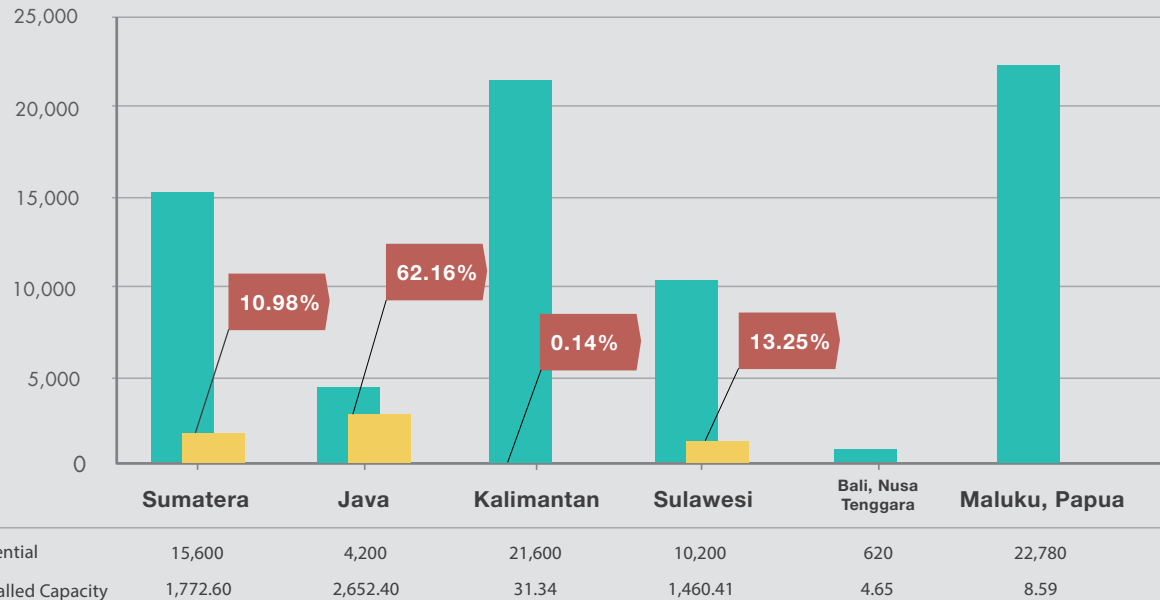
- Preliminary Survey
- Ready to be developed
- Detailed Survey
- Installed

| Island            | Total Location | Potency (MW)  |
|-------------------|----------------|---------------|
| 1. Sumatera       | 86             | 13,470        |
| 2. Jawa & Bali    | 76             | 10,013        |
| 3. Nusa Tenggara  | 22             | 1,471         |
| 4. Kalimantan     | 12             | 145           |
| 5. Sulawesi       | 56             | 2,939         |
| 6. Maluku & Papua | 33             | 1,126         |
| <b>TOTAL</b>      | <b>285</b>     | <b>29,164</b> |



# HYDRO POTENTIAL AND INSTALLED CAPACITY

(MW)

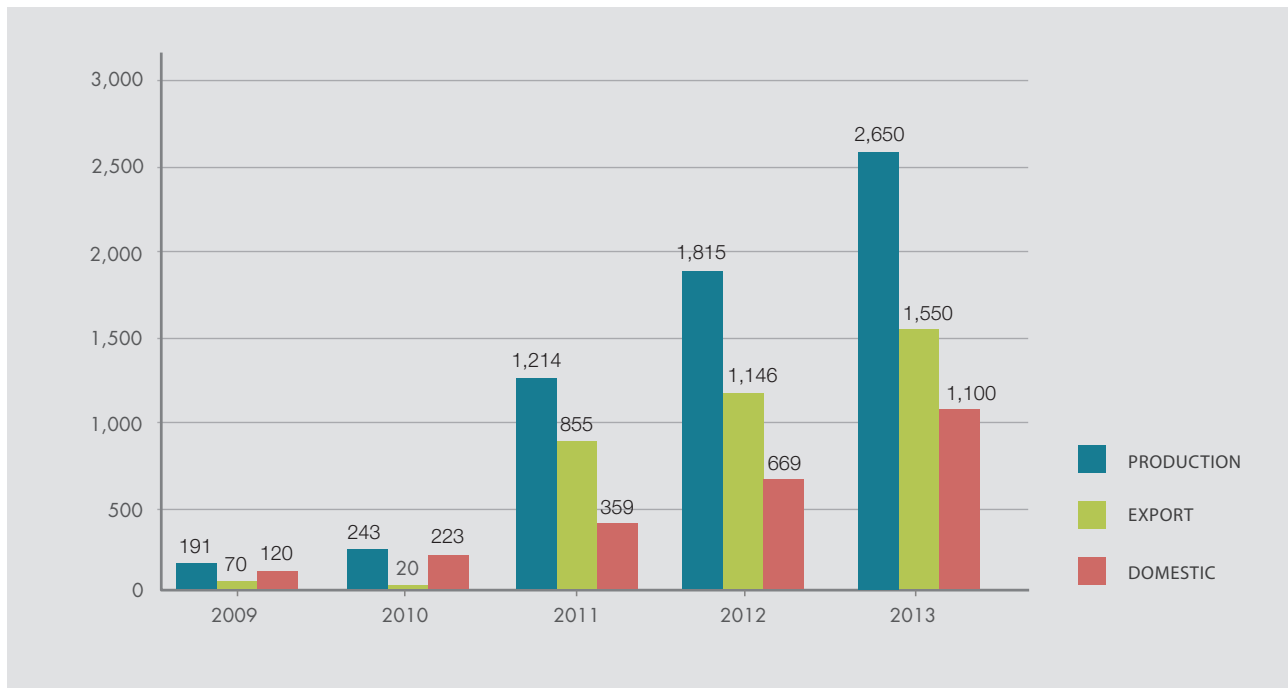


Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

- Total hydropower potential 75 GW, installed capacity 5,940MW (7.9% of potential)
- Installed capacity data included minihydro and microhydro powerplant.

# BIO-FUEL PRODUCTION

(Thousand KL)



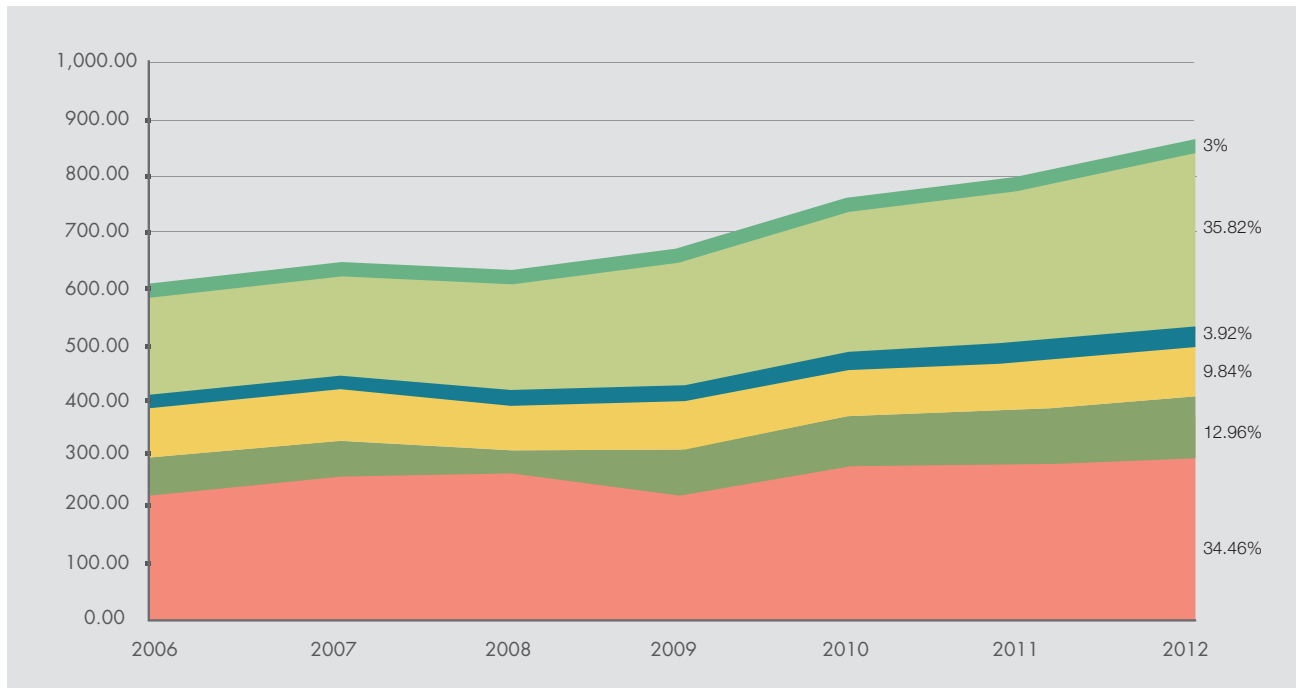
Source: Ministry of Energy and Mineral Resources, reprocessed by NEC



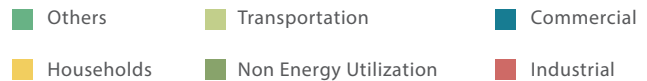
## **ENERGY CONSUMPTION BY SECTOR**

# BIO-FUEL PRODUCTION

(MBOE)

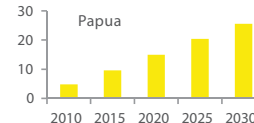
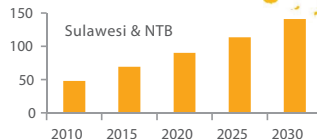
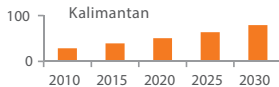
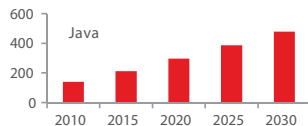
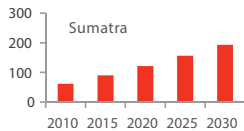
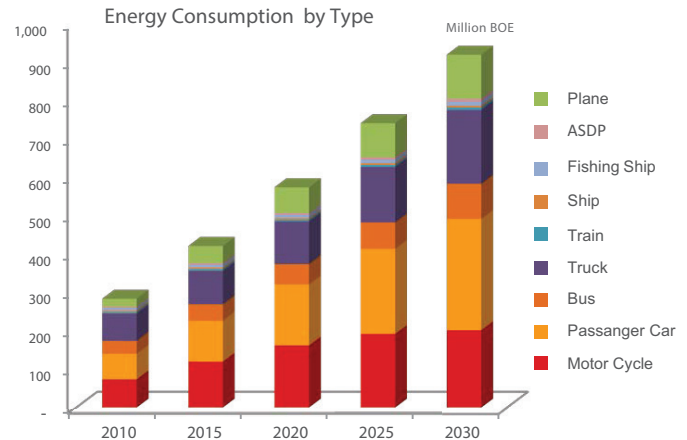


Source: Ministry of Energy and Mineral Resources, reprocessed by NEC  
Excluded Biomass



# TRANSPORTATION

| TRANSPORTATION                |  |  |  |  |  |
|-------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Capacity (people)             | 1,500                                                                             | 40                                                                                | 500                                                                               | 2                                                                                 | 1,500                                                                             |
| Consumption (Liter/Km)        | 3                                                                                 | 0.5                                                                               | 40                                                                                | 0.08                                                                              | 10                                                                                |
| Consumption (Liter/KM/People) | 0.002                                                                             | 0.0125                                                                            | 0.05                                                                              | 0.04                                                                              | 0.006                                                                             |



Source: Ministry of Transportation, reprocessed by NEC

# ENERGY CONSUMPTION VEHICLES BY TYPES

(Million BOE)



## Passenger Car

Subsidized fuel consumption for passenger cars are very huge, the total fuel subsidized by the Government almost 30% consumed by passenger cars, and 44% by motorcycles.



## Bus

Bus consumed 8% from total fuel subsidized by government.



## Truck

Truck consumed 17% from total fuel subsidized by government.



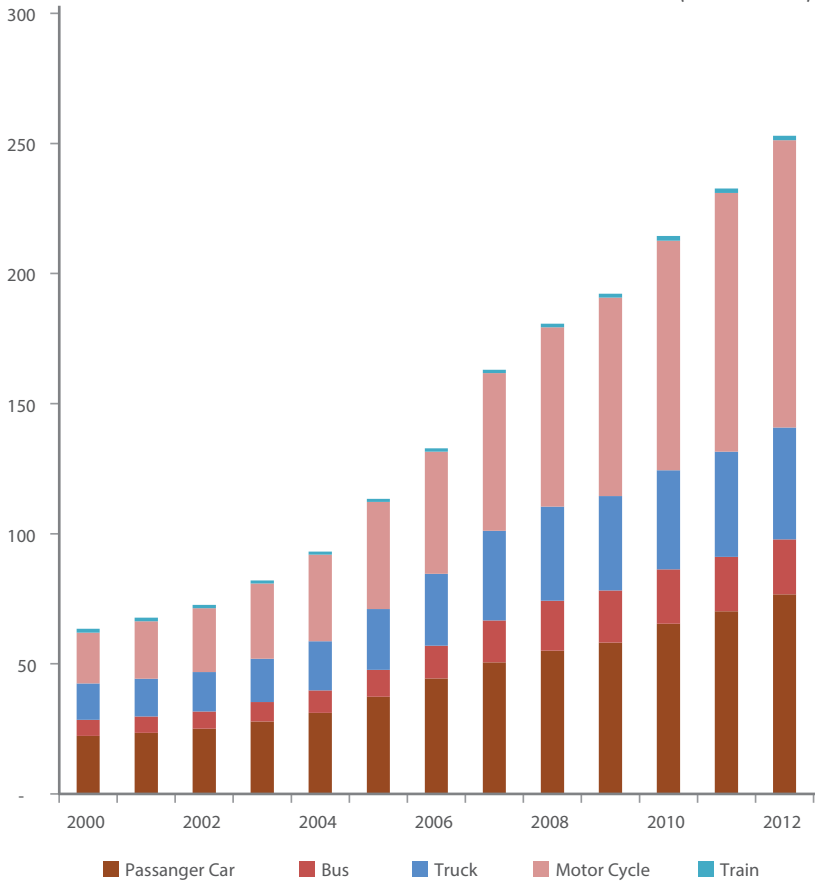
## Motor Cycle

Motor cycle growth rate is very high (15.3% per year), exceed the growth rate of total vehicles by 14% per year.

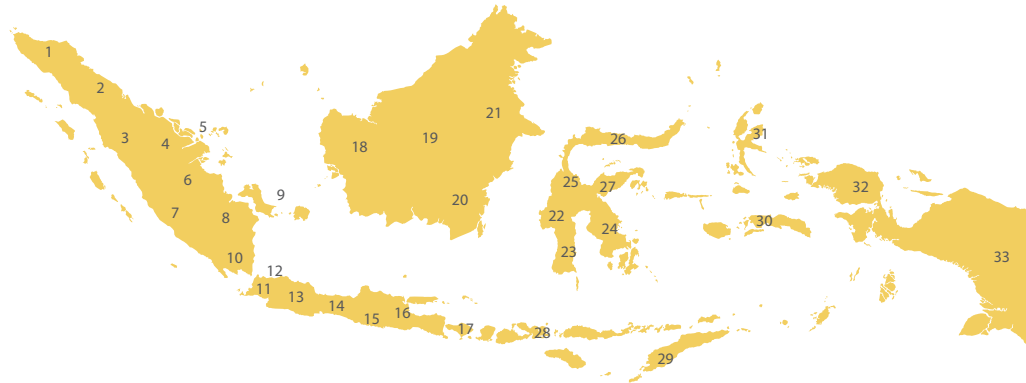


## Train

Train consumed only 1% from the total fuel. To date, trains service are only available in Java and Sumatera islands.



# ENERGY CONSUMPTION BY INDUSTRY



| 1. NAD | 2. Sumatera Utara | 3. Sumatera Barat | 4. Riau | 5. Kepulauan Riau | 6. Jambi | 7. Bengkulu | 8. Sumatera Selatan | 9. Bangka Belitung | 10. Lampung | 11. Banten | 12. DKI Jakarta | 13. Jawa Barat |
|--------|-------------------|-------------------|---------|-------------------|----------|-------------|---------------------|--------------------|-------------|------------|-----------------|----------------|
| 17588  | 132431            | 29421             | 175787  | 231052            | 177926   | 6835        | 61765               | 416092             | 83057       | 417570     | 163327          | 819689         |
| 5      | 174913            | 1010841           | 286815  | 5460              | 68770    | 0           | 206000              | 62480              | 212653      | 646144     | 41164           | 1924763        |
| 101    | 14937             | 2                 | 13503   | 5336              | 48       | 0           | 10836               | 7                  | 202         | 53399      | 15551           | 82267          |
| 16118  | 455356            | 304363            | 103517  | 1035949           | 121565   | 2771        | 119278              | 6604               | 81474       | 2224937    | 1959680         | 7852385        |

| 14. Jawa Tengah | 15. DI Yogyakarta | 16. Jawa Timur | 17. Bali | 18. Kalimantan Barat | 19. Kalimantan Tengah | 20. Kalimantan Selatan | 21. Kalimantan Timur | 22. Sulawesi Barat | 23. Sulawesi Selatan | 24. Sulawesi Tenggara | 25. Sulawesi Tengah | 26. Sulawesi Utara |
|-----------------|-------------------|----------------|----------|----------------------|-----------------------|------------------------|----------------------|--------------------|----------------------|-----------------------|---------------------|--------------------|
| 131883          | 25705             | 524510         | 9998     | 36848                | 72391                 | 66992                  | 70442                | 4071               | 39125                | 12678                 | 5056                | 39125              |
| 530599          | 131604            | 663113         | 9371     | 17357                | 53                    | 124757                 | 42805                | 3                  | 1254647              | 171477                | 1871                | 1254647            |
| 1347            | 11                | 153474         | 17       | 5                    | 30                    | 1                      | 12                   | -                  | 82                   | 3                     | 20                  | 82                 |
| 1775626         | 128090            | 3095922        | 21370    | 11718                | 3158                  | 32071                  | 90139                | 155                | 342012               | 879                   | 5398                | 342012             |

| 27. Gorontalo | 28. Nusa Tenggara Barat | 29. Nusa Tenggara Timur | 30. Maluku | 31. Maluku Utara | 32. Papua Barat | 33. Papua |
|---------------|-------------------------|-------------------------|------------|------------------|-----------------|-----------|
| 918           | 1118                    | 39125                   | 6075       | 16               | 64810           | 15285     |
| 226           | 41                      | 1254647                 | -          | -                | 599             | 39        |
| -             | 17                      | 82                      | 3          | -                | 7042            | 29        |
| 261           | 1892                    | 342012                  | 977        | 54               | 4995            | 1333      |

## Total

|                                |             |
|--------------------------------|-------------|
| Fuel Oil (KiloLiter)           | : 3,874,768 |
| Coal (Ton)                     | : 7,648,460 |
| Gas (Thousand M <sup>3</sup> ) | : 358,288   |
| Electricity (Thousand Kwh)     | : 1,984,178 |

# ENERGY DEMAND PROJECTION (2012-2017)



|                                    |             |
|------------------------------------|-------------|
| 1. KS-POSCO Cilegon Steel Industry | (700 MW)    |
| 2. Subang Hi-Tech Industry         | (600 MW)    |
| 3. Cilamaya Otomotif               | (220 MW)    |
| 4. Majalengka Textile              | (175,4 MW)  |
| 5. Kendal Textile                  | (159,12 MW) |
| 6. Demak Alumina                   | (60 MW)     |
| 7. Kulonprogo Steel                | (529,2 MW)  |
| 8. Boyolali Textile                | (56,4 MW)   |
| 9. Tuban Chemical                  | (40 MW)     |
| 10. Jombang Footwear               | (163,64 MW) |
| 11. Gresik Petrochemical           | (57 MW)     |
| 12. Lamongan Shipment              | (190 MW)    |

|                       |            |
|-----------------------|------------|
| 13. Bintuni Oil & Gas | (400 MW)   |
| 14. Tangguh Oil & Gas | (430,4 MW) |

|                         |             |
|-------------------------|-------------|
| 15. Bitung Warehouse    | (122 MW)    |
| 16. Palu Rattan         | (300 MW)    |
| 17. Takalar Ferronickel | (1.000 MW)  |
| 18. Gowa Cocoa          | (168,42 MW) |
| 19. Jeneponto Salt      | (170 MW)    |

|                                     |          |
|-------------------------------------|----------|
| 20. Halmahera Timur Stainless Steel | (250 MW) |
|-------------------------------------|----------|

|                         |            |
|-------------------------|------------|
| 21. Kariangau Oil & Gas | (397,9 MW) |
| 22. Batu Licin Steel    | (500 MW)   |
| 23. Landak Rubber       | (400 MW)   |
| 24. Tayan Alumina       | (400 MW)   |

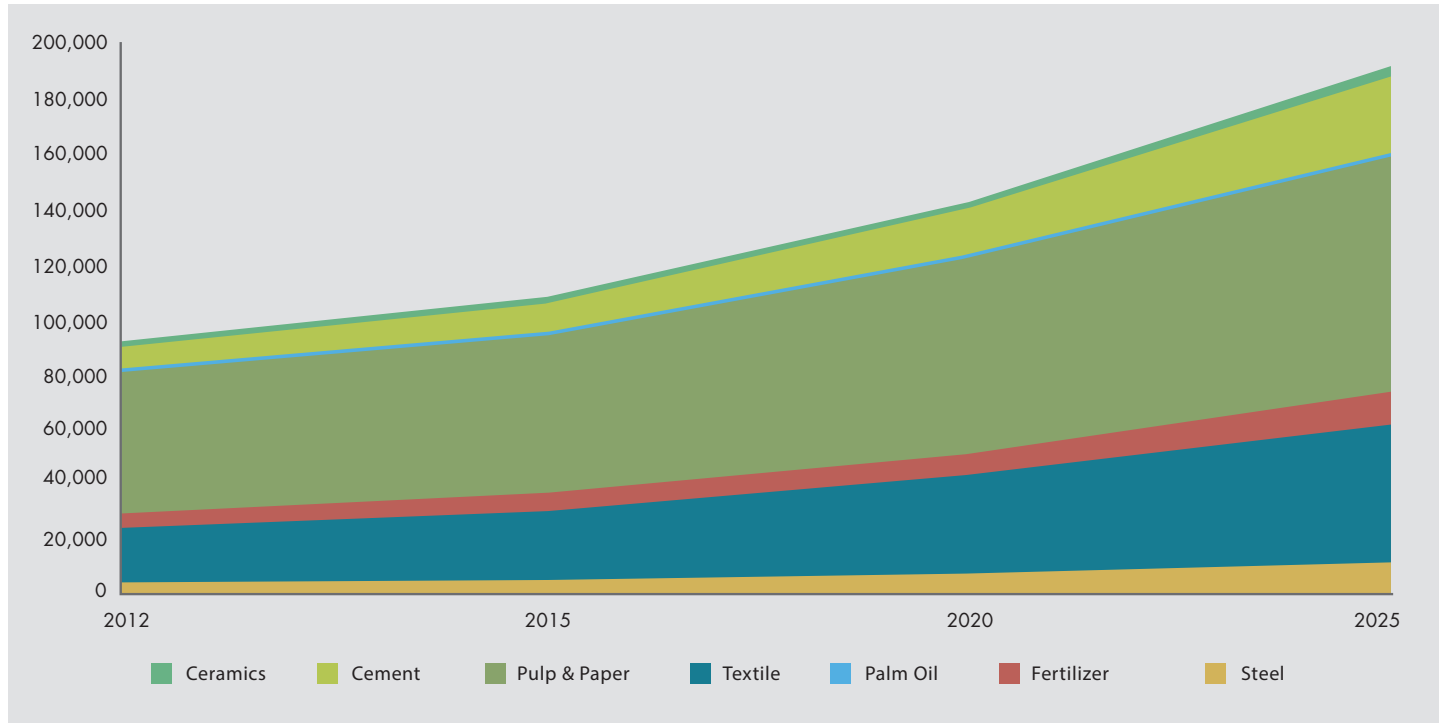
|                                       |             |
|---------------------------------------|-------------|
| 25. Bangka Tin                        | (153,08 MW) |
| 26. Tanjung Api-Api Coal Gasification | (808,92 MW) |
| 27. Muara Enim Rubber                 | (120 MW)    |
| 28. Tanggamus Maritim                 | (700 MW)    |
| 29. Tanjung BÜton Oleochemical        | (1.000 MW)  |
| 30. Kuala Tanjung Alumina             | (200 MW)    |
| 31. Sei Bamban Rubber                 | (22,5 MW)   |
| 32. Sei Mangke Palm Oil               | (400,4 MW)  |

Total : 10,864,38 MW



# ENERGY DEMAND FOR INDUSTRIAL SECTOR

(GWh)



Source: Ministry of Industry and BPS, reprocessed by NEC

Beyond the need for feedstock in the fertilizer industry, the pulp and paper industry is the largest user of energy i.e. by 59.3 percent. The cement industry is projected to grow most rapidly, from 6.5 percent in 2012 to 12.4 percent in 2025

# POWER PLANT POTENTIAL BASED ON PALM OIL WASTE

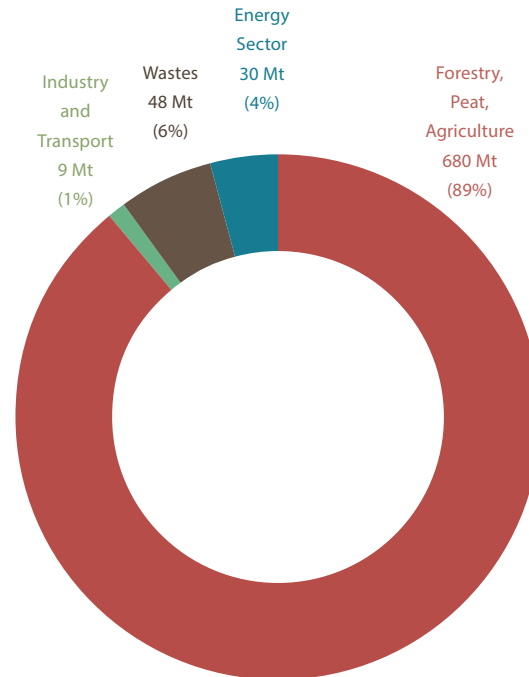
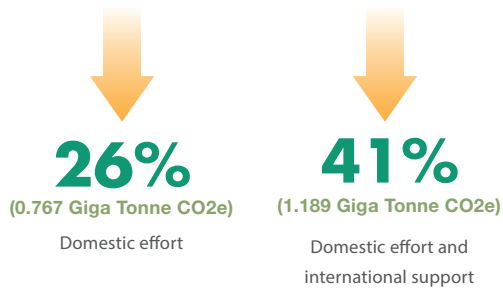




**CO<sub>2</sub> EMISSIONS**

# NATIONAL COMMITMENT TO REDUCE GHG

Presidential Regulation Number 61 Year 2011  
to reduce greenhouse gasses (GHG) emission by 2020  
(Indonesia Commitment in G-20 Pittsburgh and COP15)



BP Statistical Review, reprocessed by NEC

# EMISSION REDUCTION ROAD MAP IN ENERGY SECTOR

## ACTION PLAN

Million Ton CO<sub>2</sub>

**2010-2014**

- Energy Audit
- Energy Saving Lamps
- The provision and management of NRE
- Biogas utilization

- Gas for public transportation
- Increasing gas city connection to households
- Developing the LPG Mini Refinery Plant
- Reclamation for the ex-mining location

**5.06**

**2015-2020**

- Energy Audit
- Energy Saving Lamps
- The provision and management of NRE
- Biogas utilization

- Gas for public transportation
- Increasing gas city connection to households
- Developing the LPG Mini Refinery Plant
- Reclamation for the ex-mining location

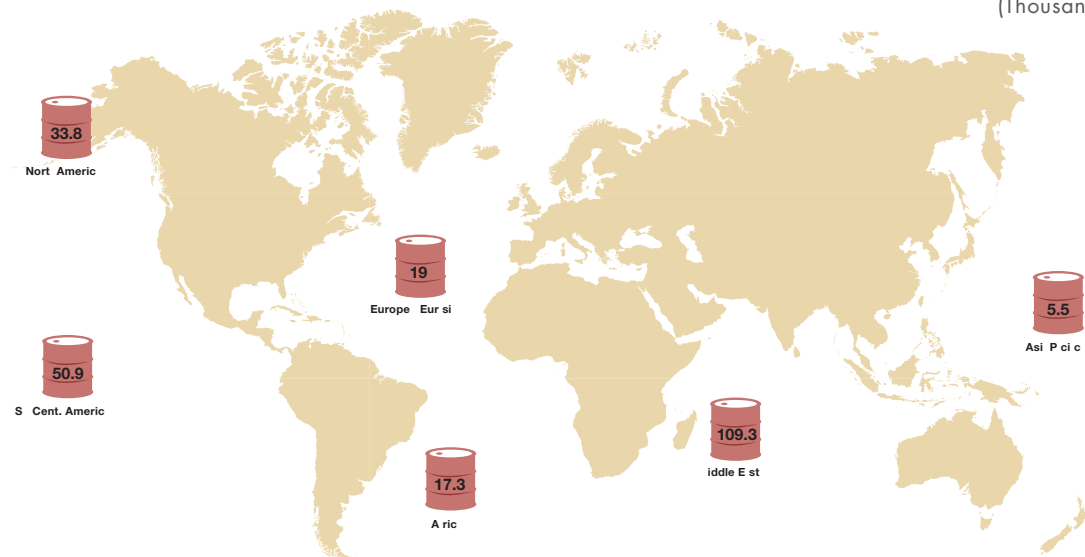
**18.08**

A solid orange horizontal bar spanning the width of the slide, positioned to the left of the title text.

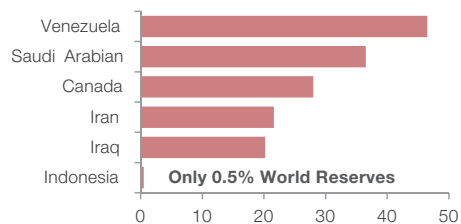
## **INDONESIA'S POSITION IN THE GLOBAL CONTEXT**

# OIL

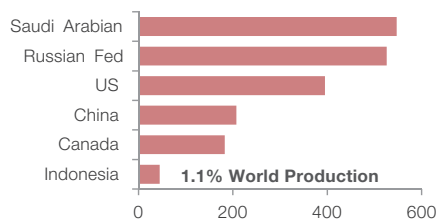
(Thousand Million Tonnes)



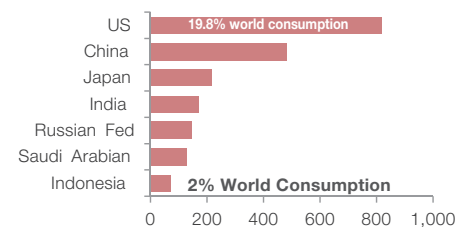
Proven Oil in selected countries 2012



Production Oil in selected countries 2012

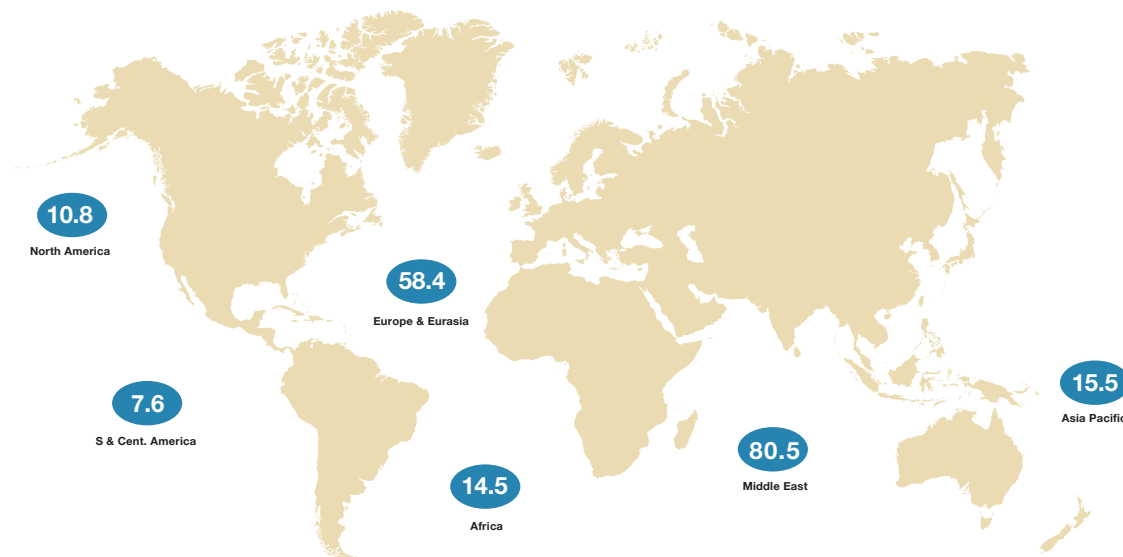


Consumption Oil in selected countries 2012

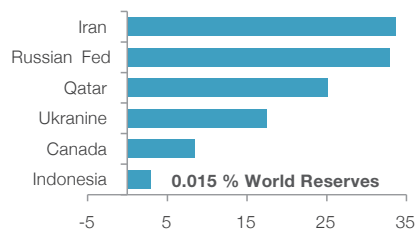


# GAS

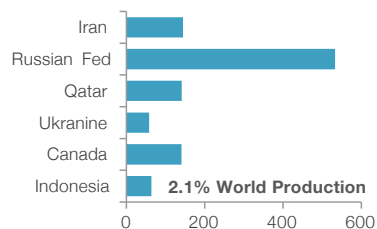
(TCM)



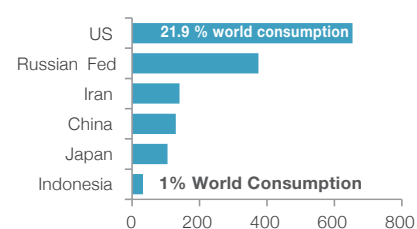
**Proven Natural Gas in selected countries, 2012**



**Production Gas in selected countries, 2012 (MTOE)**



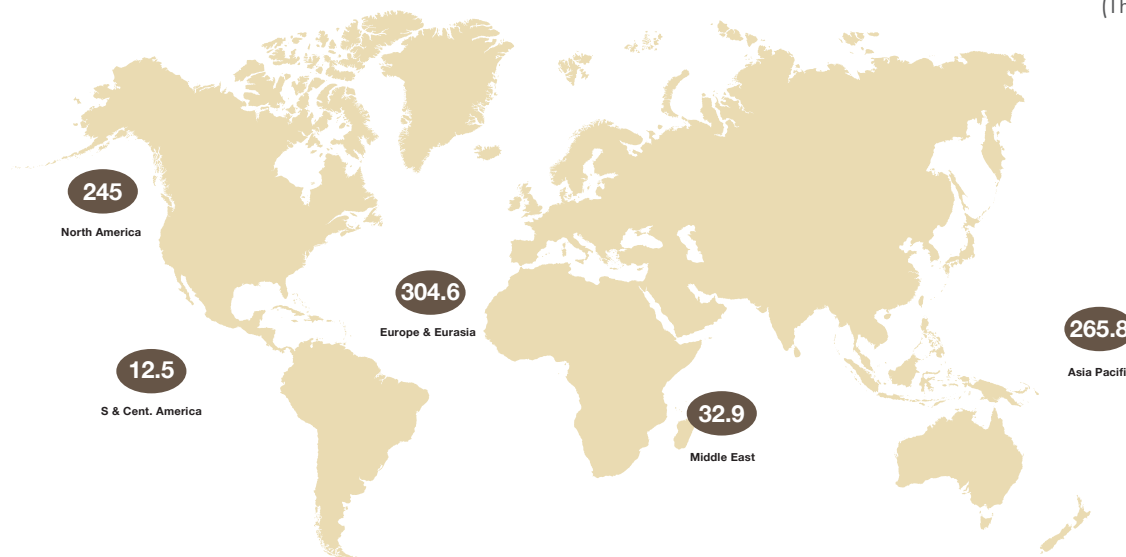
**Consumption Gas in selected countries, 2012 (MTOE)**



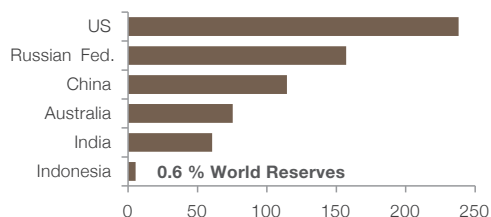


# COAL

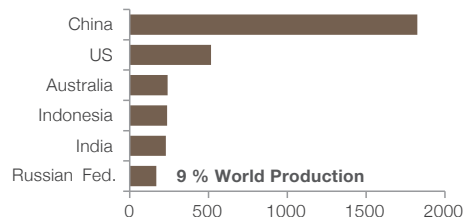
(Thousand MT)



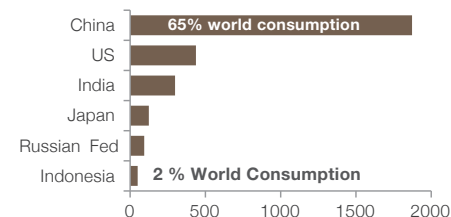
Proven Coal in selected countries, 2012



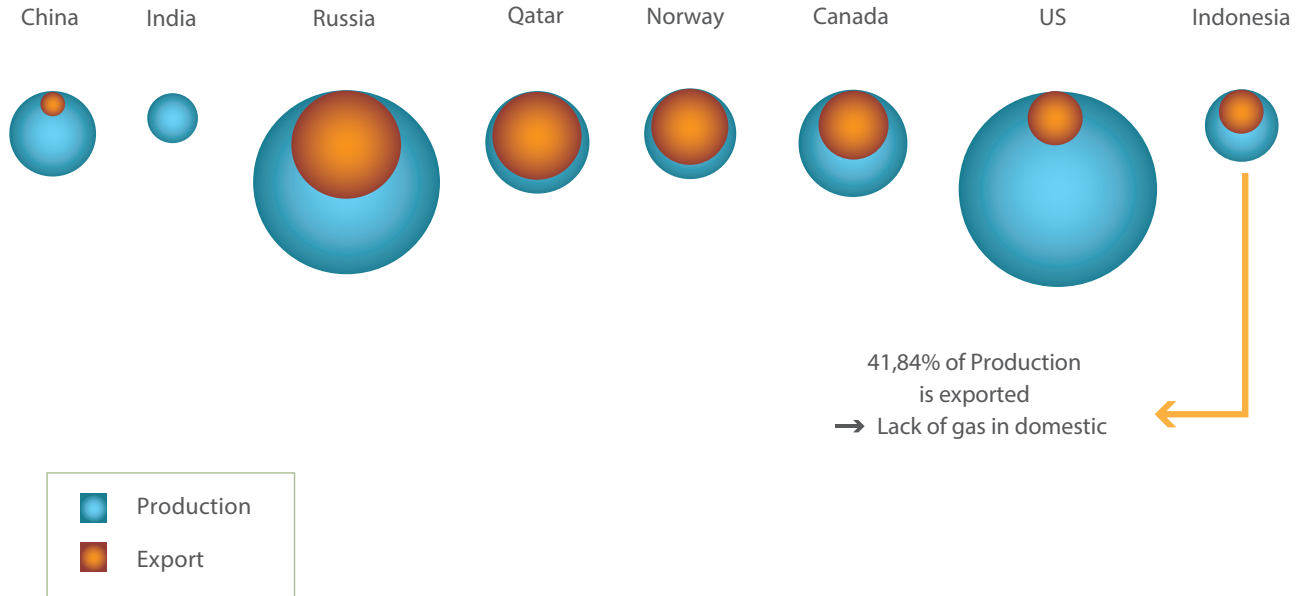
Production Coal in selected countries, 2012 (MTOE)



Consumption Coal in selected countries, 2012 (MTOE)

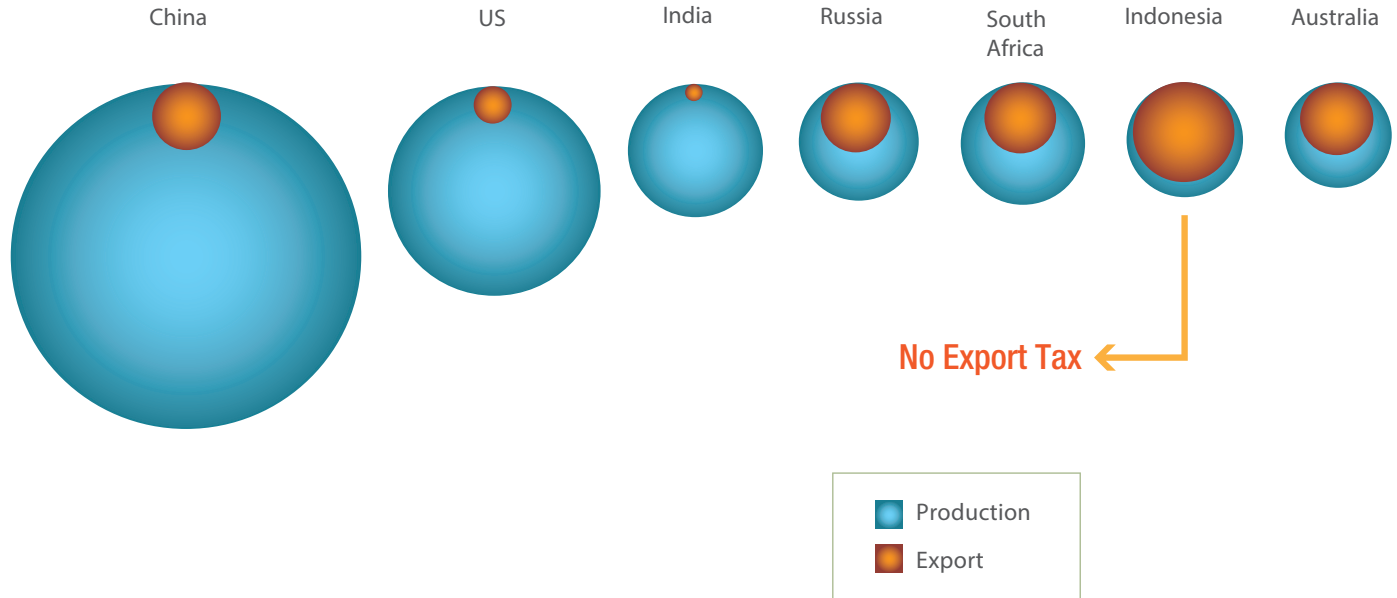


# GAS PRODUCTION-EXPORT COMPARISON



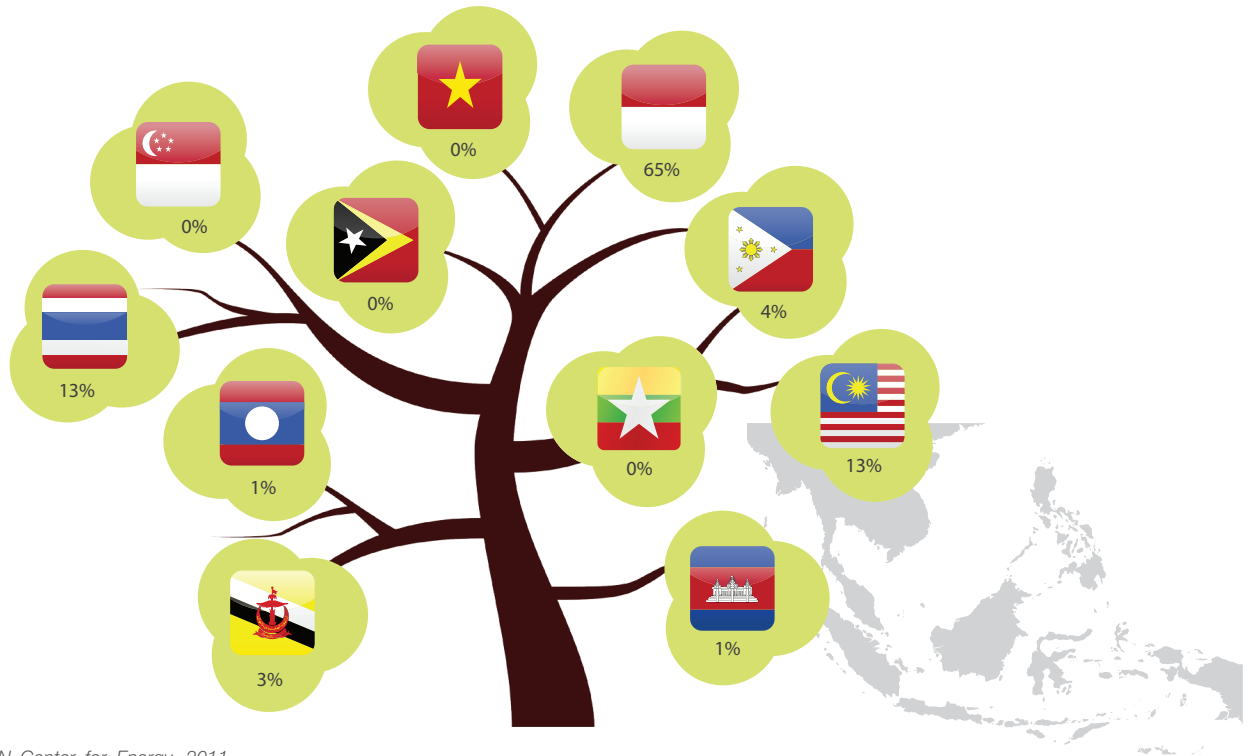
BP Statistical Review, reprocessed by NEC

# COAL PRODUCTION-EXPORT COMPARISON



Source: Wood Mackenzie coal supply service, ANZ, reprocessed by NEC

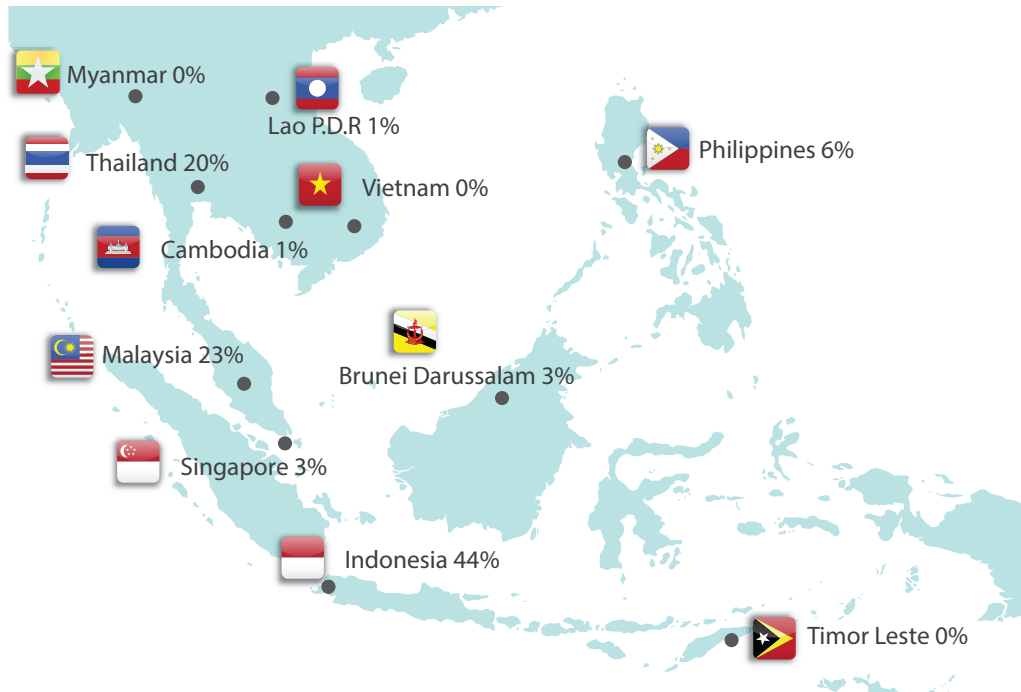
# SHARE OF ASEAN PRIMARY ENERGY PRODUCTION



Source: ASEAN Center for Energy, 2011

Note : Data of Myanmar and Vietnam are not yet provided

# SHARE OF ASEAN ENERGY DEMAND



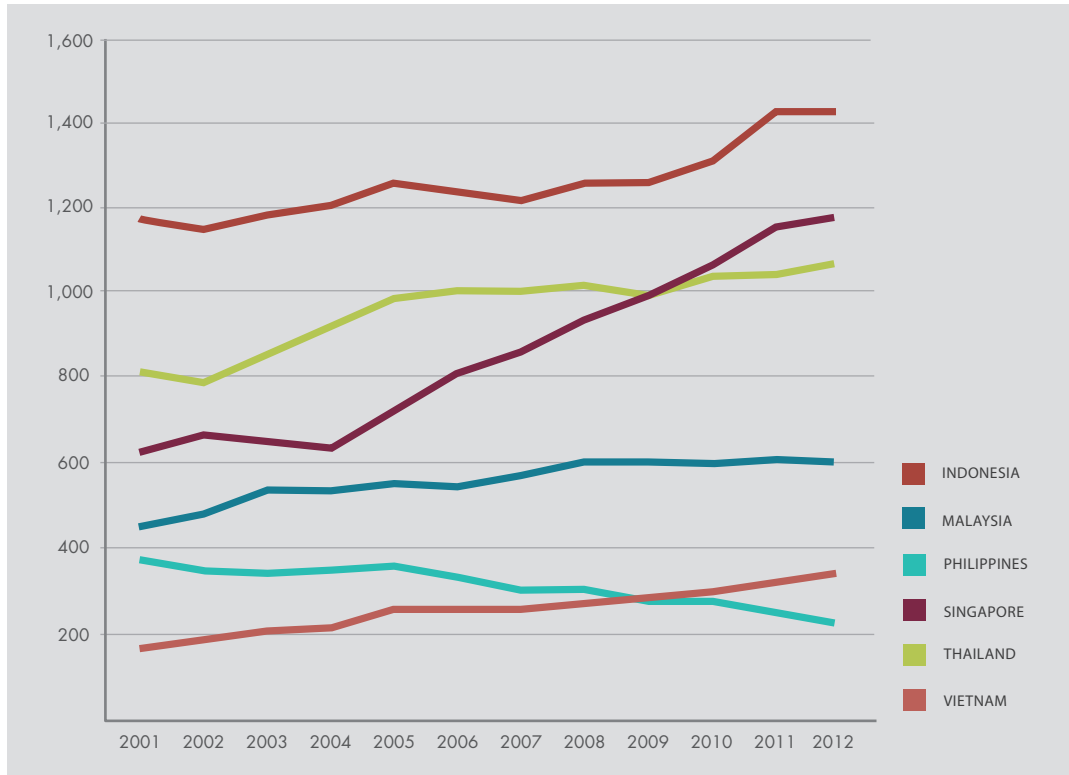
- Southeast Asia's energy demand increases by over 80% in 2013 –2035 (sesuaikan kajian dari ACE)
- Oil demand rises from 4.4 mb/d today to 6.8 mb/d in 2035, natural gas demand increases by 80% to 250 bcm

Source : ASEAN Center for Energy, 2011

Note : Data of Myanmar and Vietnam are not yet provided

# ASEAN OIL CONSUMPTION

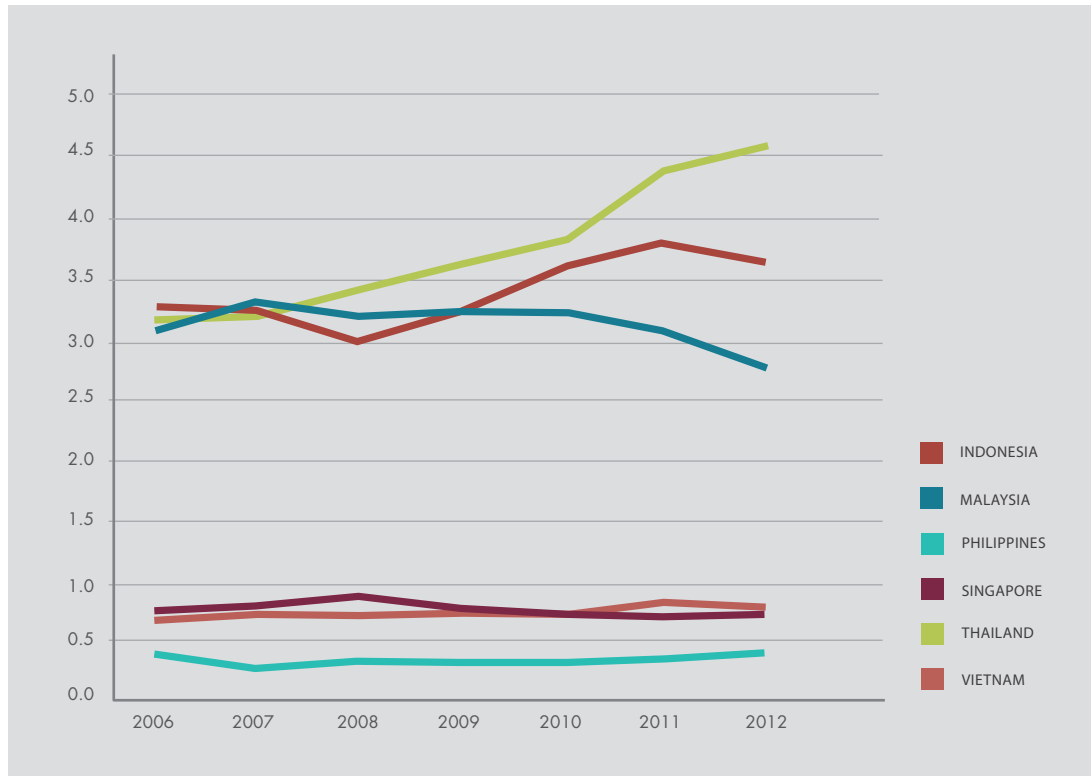
(Thousand Barrel per Day)



Source: Data of other ASEAN member states are not yet provided

# ASEAN GAS CONSUMPTION

(Thousand Barrel per Day)



Source: Data of other ASEAN member states are not yet provided

# ABBREVIATION

---

| Abbreviation |                                   |
|--------------|-----------------------------------|
| <b>Bbl</b>   | Barrel                            |
| <b>BOE</b>   | Barrels of Oil Equivalent         |
| <b>BOEPD</b> | Barrels Of Oil Equivalent Per Day |
| <b>GDP</b>   | Gross Domestic Product            |
| <b>GW</b>    | Giga Watt                         |
| <b>GWh</b>   | Giga Watt Hour                    |
| <b>KL</b>    | Kilo Litter                       |
| <b>LPG</b>   | Liquefied Petroleum Gas           |
| <b>MBOE</b>  | Million Barrels of Oil Equivalent |
| <b>MMSTB</b> | Million Stock Tank Barrels        |
| <b>MMSCF</b> | Million Standard Cubic Feet       |
| <b>MT</b>    | Million Ton                       |
| <b>MTOE</b>  | Million Ton of Oil Equivalent     |
| <b>MW</b>    | Mega Watt                         |
| <b>MWh</b>   | Mega Watt Hour                    |
| <b>KL</b>    | Kilo Litter                       |
| <b>KWh</b>   | Kilo Watt Hour                    |







**DEN**

**NATIONAL ENERGY COUNCIL**

Gatot Subroto Kav. 49 4th Floor,  
South Jakarta 12950 Indonesia

Telp: +62 21 5292 1621

Fax: +62 21 5292 0190

[www.den.go.id](http://www.den.go.id)